

Introduction to Multimodal Deep Learning

Large Foundational Models and Physical World Models for Multimodal AI

Lecture 1

Jin-Hwa Kim and Sangdoo Yun

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Class Introduction

Course objectives

- Understand the design principles of multimodal foundation model theories and applications: architecture, training recipe, data, and evaluation.
- Understand the design principles of multimodal generative model theories and applications *focusing on 3D/4D and video generation*.
- Critically review and discuss the emerging literature on multimodal foundation models and physical world models.
- Main question of the class:
"How is AI moving from *seeing and talking about the world* to *imagining and acting in it*?"

Co-lecturers



Sangdoo Yun

*Head of Future AI Center
Head of NAVER AI Lab
Guest Associate Professor at SNU AIIS*



Jin-Hwa Kim

*Leader at NAVER AI Lab
Guest Associate Professor at SNU AIIS
NAVER World Model*

General information

- Co-lectured by Sangdoo Yun & Jin-Hwa Kim
- Every Friday 1 PM from Sep 4 to Dec 18 (schedule table in the next slide)
- In Room **106***, Building 302
- Office hour: Friday 10 AM 942-208 (AIIS)
 - Sangdoo Yun (sangdoo.yun@navercorp.com)
 - Jin-Hwa Kim (jnhwkim@snu.ac.kr)
 - *Reservation only*
- Teaching assistants
 - Juno Kim (jokim@snu.ac.kr)

Schedule

Week	Date	Topic	Note
1	Sep 4	Introduction	
2	Sep 11	Multimodal Foundation Model 1	Sangdoo Yun
3	Sep 18	Multimodal Foundation Model 2	Sangdoo Yun
4	Sep 25	Multimodal Foundation Model 3 (Online)	Sangdoo Yun
5	Oct 2	Multimodal Generation 1	Jin-Hwa Kim
6	Oct 9	Multimodal Generation 2 (Online)	Jin-Hwa Kim
7	Oct 16	Multimodal Generation 3	Jin-Hwa Kim
8	Oct 23	Paper Review and Discussion Session 1	
9	Oct 30	Paper Review and Discussion Session 2	
10	Nov 6	Paper Review and Discussion Session 3	
11	Nov 13	Paper Review and Discussion Session 4	
12	Nov 20	Paper Review and Discussion Session 5	
13	Nov 27	Paper Review and Discussion Session 6	
14	Dec 4	Paper Review and Discussion Session 7	
15	Dec 11	NeurIPS 2026 (No Class)	
16	Dec 18	Paper Review and Discussion Session 8	

Course format

- 6 sessions – Instructor lectures
 - Multimodal Foundation Models, 3 sessions (*Sangdoo Yun*)
 - Multimodal Generation, 3 sessions (*Jin-Hwa Kim*)
- 8 sessions – **Student presentations and discussion**
 - Each session consists of 3~4 presentation sets.
 - One set: 20-min presentation + 15-min discussion + transition (**≈ 40 min**)
 - 3 sets $\approx$ 120 minutes, 4 sets $\approx$ 160 minutes
 - Total class slot: 170 minutes, including a short break

Course format – Student presentation

- Paper assignment begins: September 14
 - *Every student presents one paper during this course*
 - Students bid papers from 50-60 papers curated by the instructors, and presentation dates
- One 20-minute conference-style paper talk
 - Not a sequence of summary slides
 - The presentation should explain:
 - *Problem & significance → Essential background → Technical core → Evidence & its limits → Your critique & open questions*
 - You should be able to explain and defend every claim, equation, figure, and conclusion included in your presentation.
 - Final slide deck should be uploaded at ETL before the presentation (*deadline: D-2 23:59 KST*)
-  How to Give a Great Research Talk (Simon Peyton Jones)

Course format – Discussions

- Each 20-minute presentation is followed by a 15-minute discussion with two assigned panelists.
 - *Every student participates as a panel twice during this course*
 - Panel assignments begin after presenter–paper assignments are finalized.
- The panel has two responsibilities
 - **As a reviewer:** Examine what the paper establishes and does not establish, methodological weaknesses, missing baselines or ablations, and claims that outrun the evidence.
 - **As a research collaborator:** Identify promising follow-up directions with the presenter, turn the best direction into a testable research question, etc.
- **Panel report:** Submit a short report (1-2 pages) covering a summary of the presented work, the substance of the discussion, and the follow-up research directions. (*Deadline: D+7 23:59 KST*)

Preliminaries

- Since student presentations and discussion drive this course, class time will assume the prerequisite material below.
 - Basic knowledge on deep learning, probability, linear algebra
 - Understanding Transformers architectures
 - Recommended: [Introduction to Transformers](#) (Stanford CS25)
 - Optional deep dive: [Andrej Karpathy — Let's build GPT: from scratch, in code, spelled out](#)

AI Use Policy

- Every deliverable must include a brief AI Use Note.
 - If none was used, write: “No generative AI used.”
 - State the tool/model, purpose, and prompt (optional)
- Verify every citation, number, equation, figure, and claims against the original paper or another reliable primary source.
- No live generative-AI assistance during presentations, Q&A, or panel discussions, except as an approved accommodation.

Grade

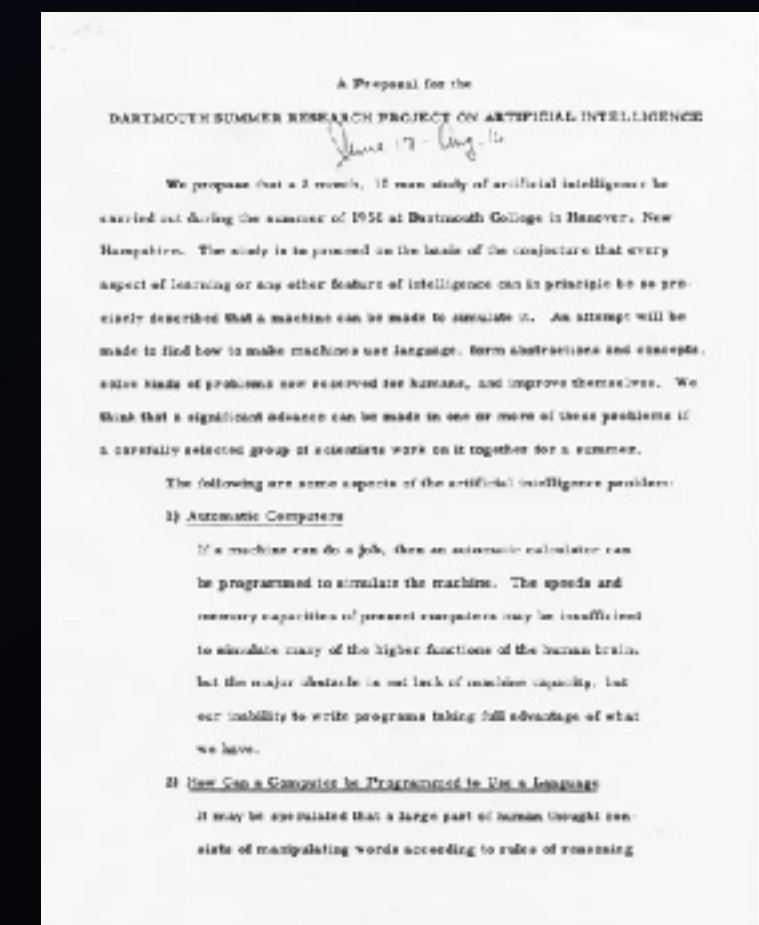
- Attendance (10%)
 - Students who are absent for over 1/3 of the class will receive a grade of F for the course. (Exceptions can be made when the course instructor deems the cause of absence unavoidable.)
- Attitude (10%)
 - Asking a thoughtful question (as determined by the instructor or invited speaker) earns 2 points.
- Presentation (50%)
 - Rubrics will be updated on class webpage
- Panel Discussion and report (30%)
 - Rubrics will be updated on class webpage

The birth of AI field

The summer of 1956 at Dartmouth

- In the summer of 1956, young scholars gathered at Dartmouth College in Hanover, New Hampshire, the founding event of *artificial intelligence as a field*.
- “The study is to proceed on the basis of the conjecture that every *aspect of learning* or any other *feature of intelligence* can in principle be so precisely described that a machine can be made to simulate it.” (McCorduck* 1979, p.93)

Marvin Minsky, Claude Shannon, Ray Solomonoff and others at the Dartmouth
(Photo: Margaret Minsky)



*Pamela McCorduck, a historian of artificial intelligence

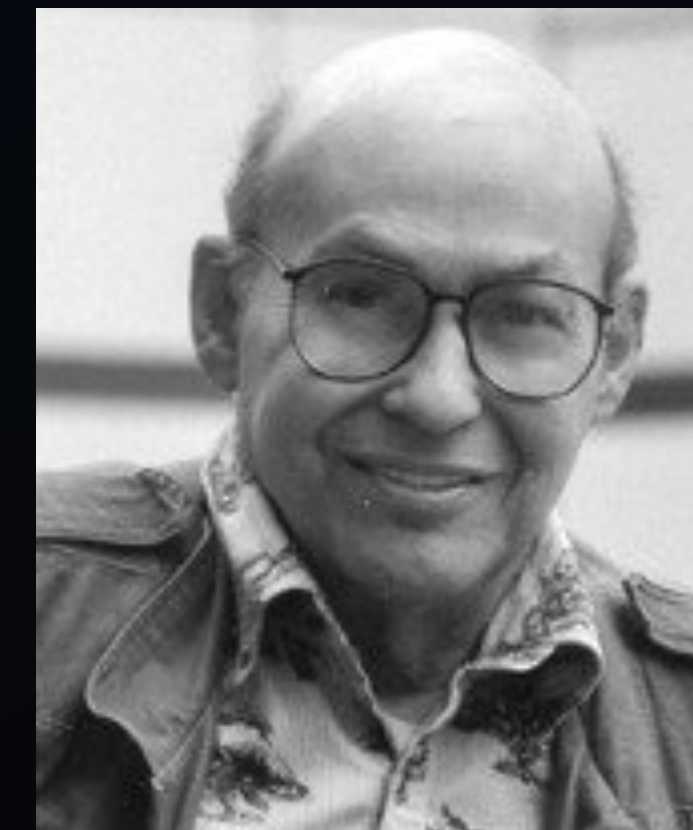
Early AI contributors

- John McCarthy (1927-2011)
 - Then, an assistant professor of mathematics at Dartmouth and eventually
 - the founder and first director of the AI Labs at both MIT (1957) and Stanford (1963)
 - the major coiner of the term *artificial intelligence (AI)*,
 - developed the programming language family *LISP* and invented *garbage collection*.
 - 1971 Turing Awardee.



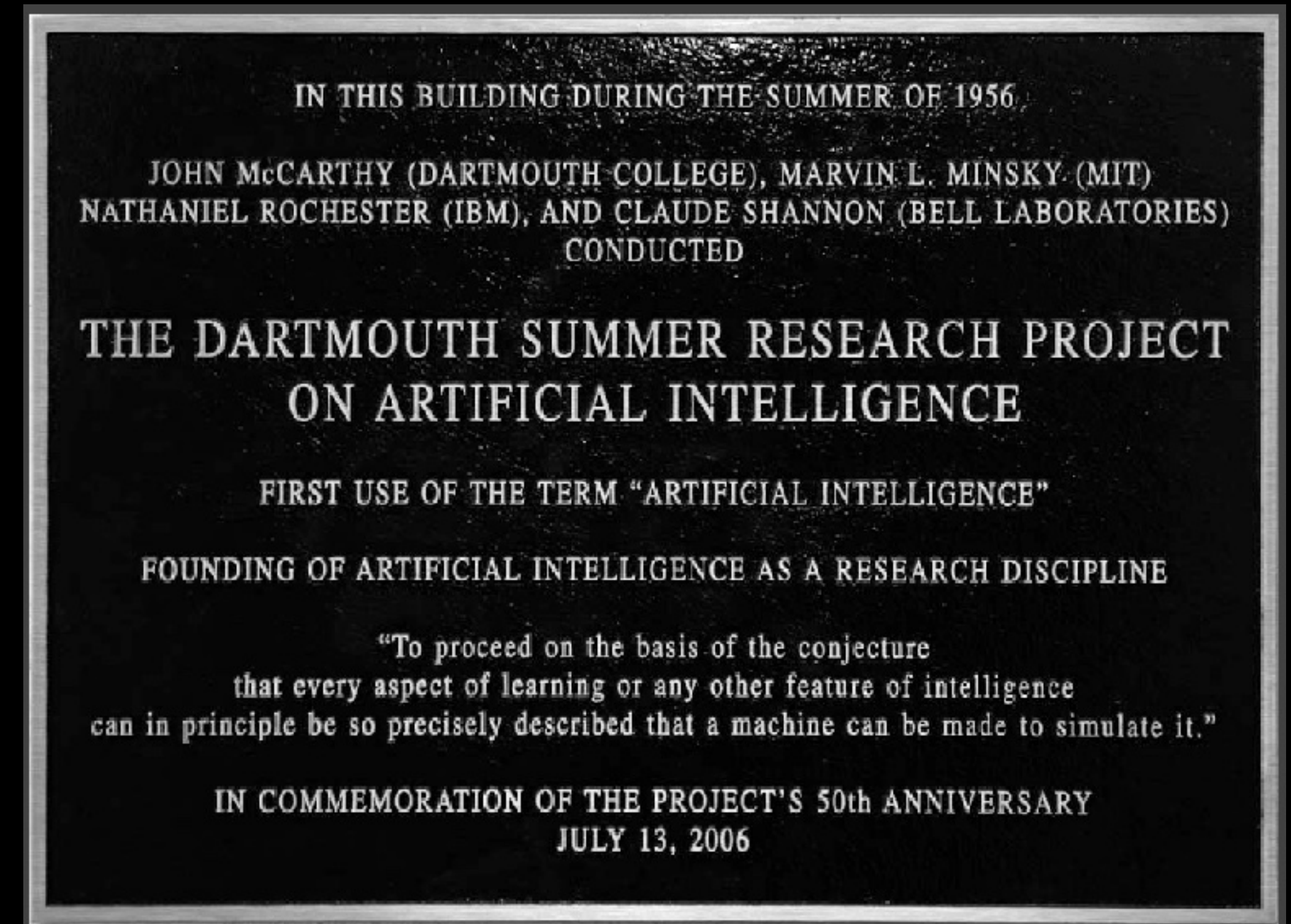
Early AI contributors

- Marvin Minsky (1927-2016)
 - Then, he was a junior fellow in mathematics and neurology at Harvard and eventually
 - a co-founder of MIT's AI Lab, and the director of the AI Lab at MIT
 - 1969 Turing Awardee.
 - The co-authored book *Perceptrons* attacked *Frank Rosenblatt's* work, becoming the seminal work in the analysis of artificial neural networks.
 - It was crucial in discouraging neural network research in the 1970s and contributing to the so-called "*AI winter*."



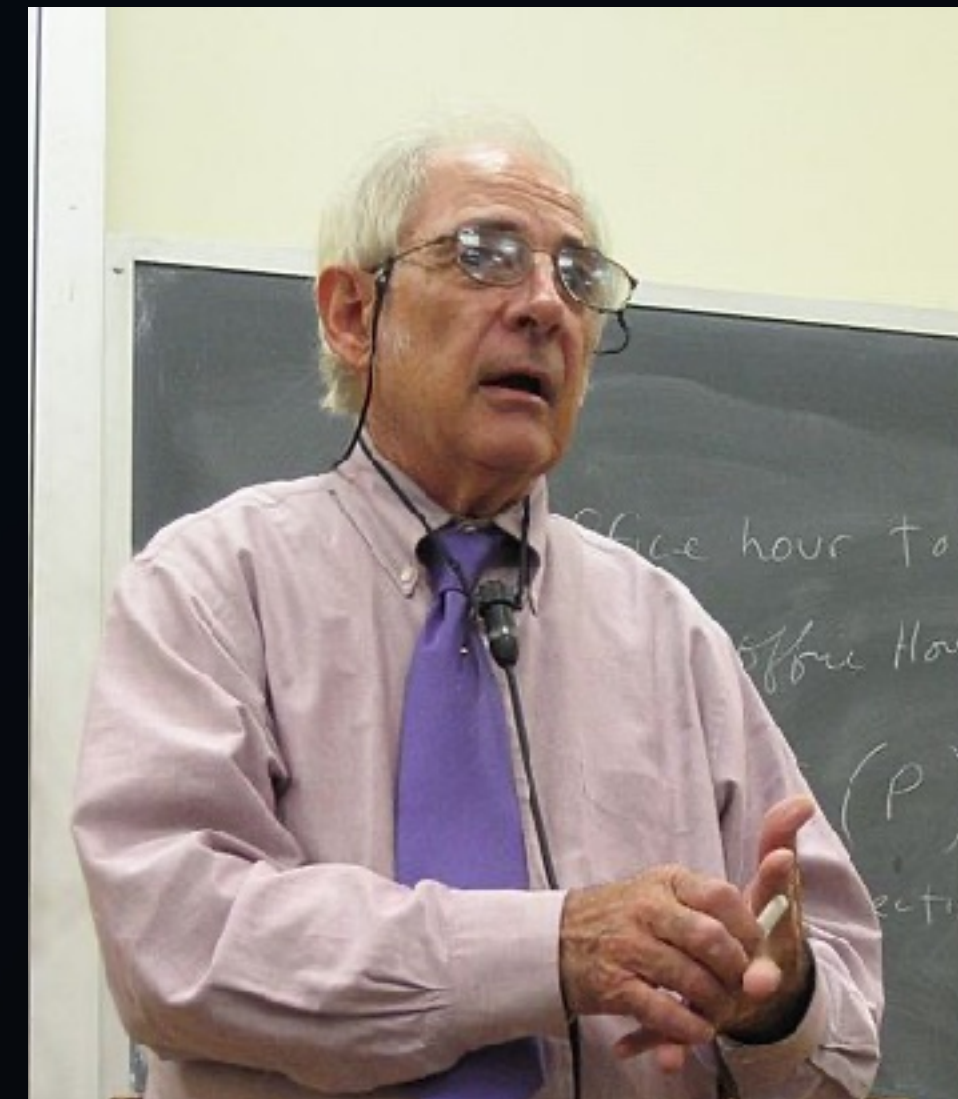
Meaning of Dartmouth

- Despite of limited success, that was symbolic advance from an older generation — Norbert Wiener, John von Neumann, Warren McCulloch, and *Alan Turing* — where in doubt in the development of AI, more than electronic computers.
- *“While no single event can lay claim to signaling the **birth** of all of cognitive science, the workshop at Dartmouth is the chief contender within the field of AI.”*



Conflicting ideas

- Strong vs. weak AI
 - “The appropriately programmed computer really *is* mind”? (John Searle, 1980)
- Generalists vs. experts
- “The dramatic promise of a *thinking machine*, the battles about the scientific status of AI have been particularly vehement.”



Photograph of John Searle by Matthew Breindel

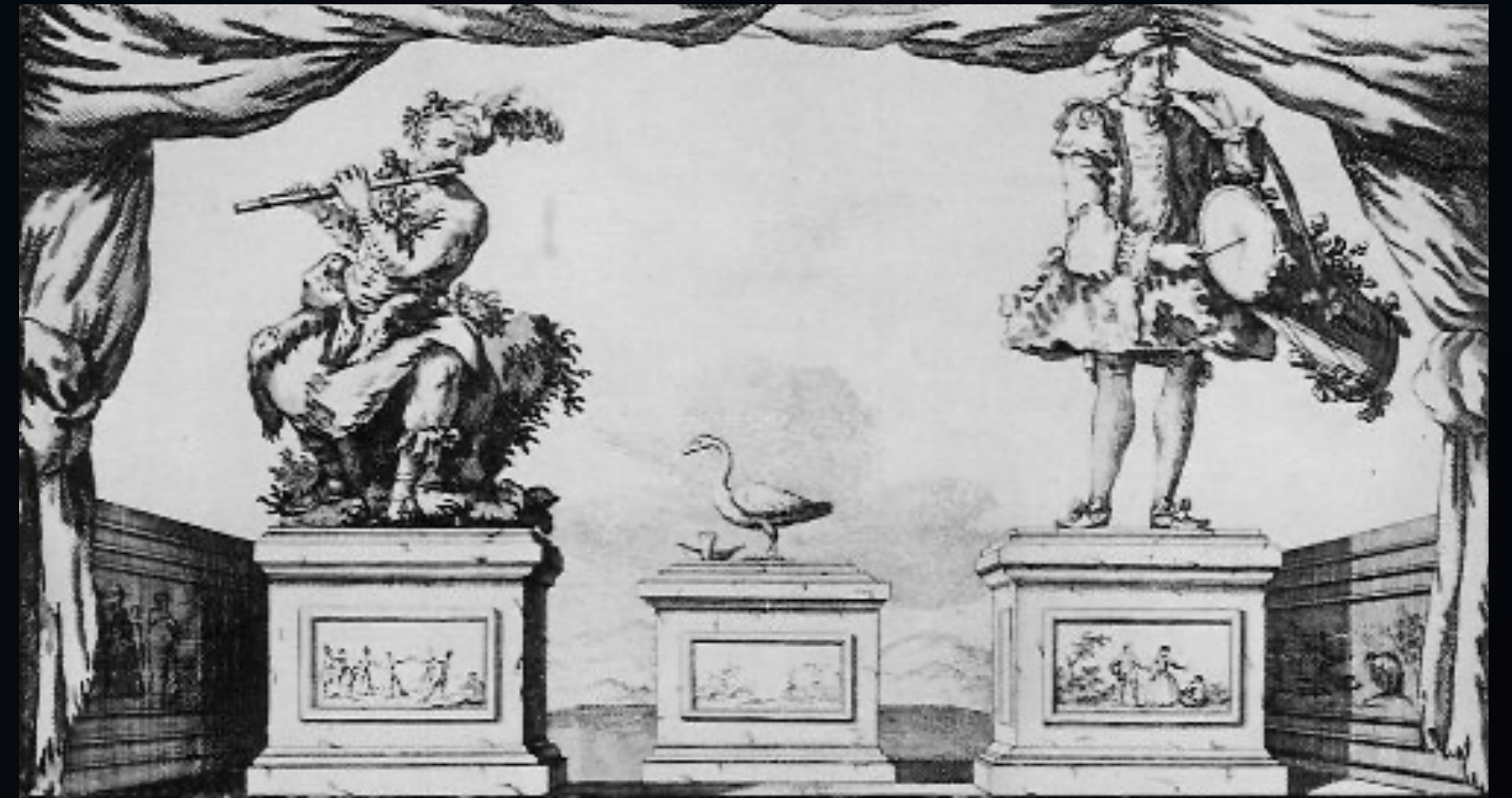
Dream of AI 🇫🇷

- René Descartes (1596-1650) was interested in the automata simulating human body, while skeptical about simulating the mind.
- L'Homme Machine (La Mettrie, 1747)
 - “the human body is a machine that winds up its own springs” (McCorduck, 1979)



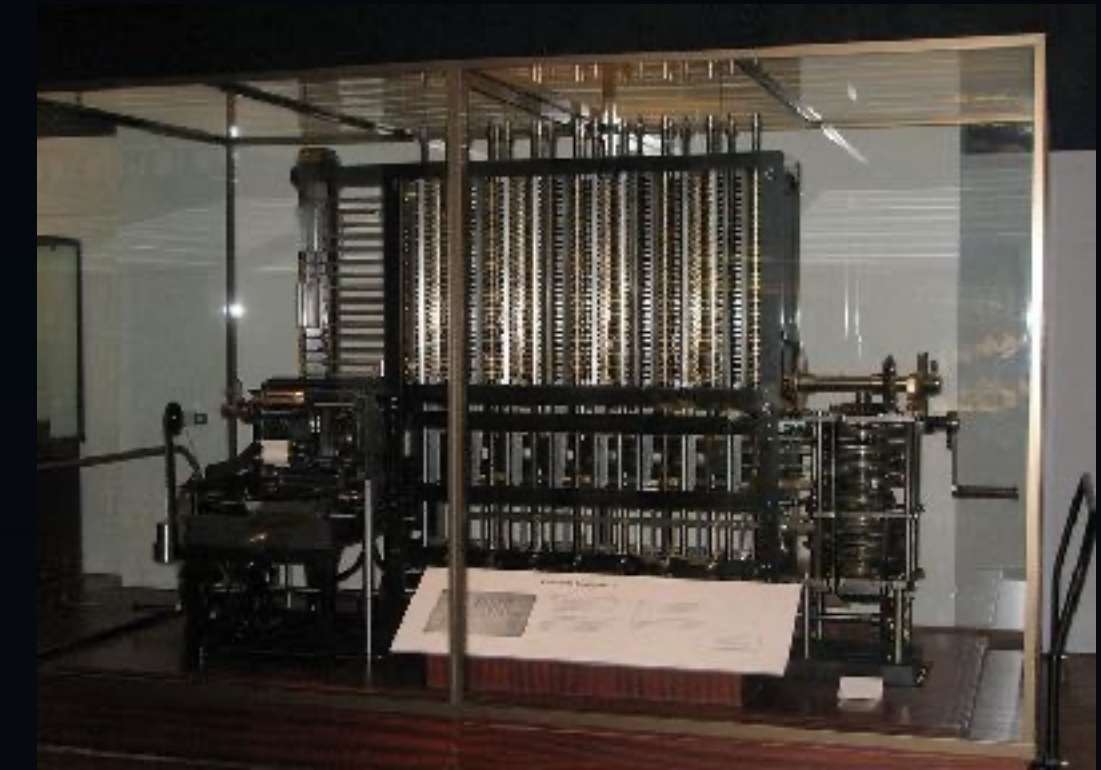
The father of simulation 🇫🇷

- Jacques de Vaucanson (1709–1782) was an automata builder who thrilled Europe with mechanical flute players, ducks, and pipe and taber players.
- Vaucanson “was concerned to formulate and validate — in the most precise and formal language available to him — a theory of the German flute player” (Fryer and Marchall, 1979)



British ambitions

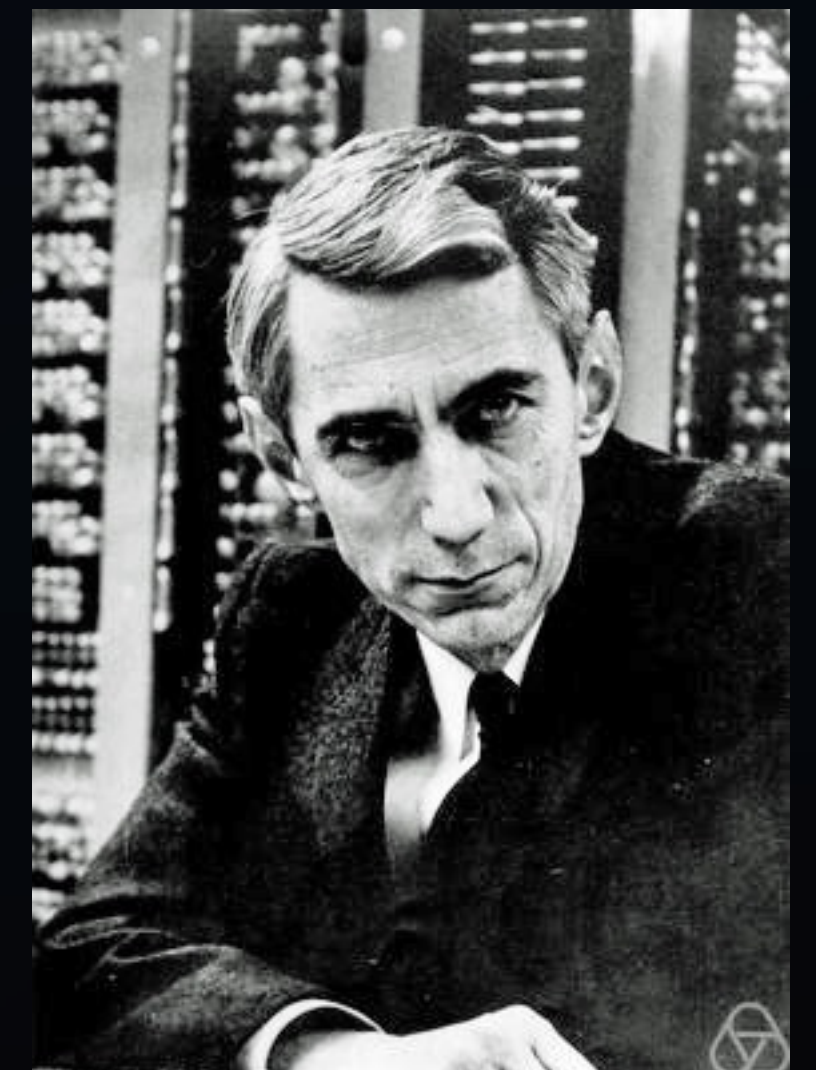
- Charles Babbage (1791–1871)
 - Automatic table calculator for any arithmetic problem, *“difference machine”*
- George Boole (1815–1864)
 - Invent basic laws of thought on the principles of *logic*, *“mental algebra”*
 - Used a set of *symbols* (e.g., a, b, x, y) to stand for the components of thought
 - *“a successful attempt to express logical propositions by symbols, the laws of whose combinations should be founded upon the laws of the mental processes which they represent, would, so far, be a step toward the philosophical language”* (Boole, 1847, p.5)



The father of information theory 🇺🇸

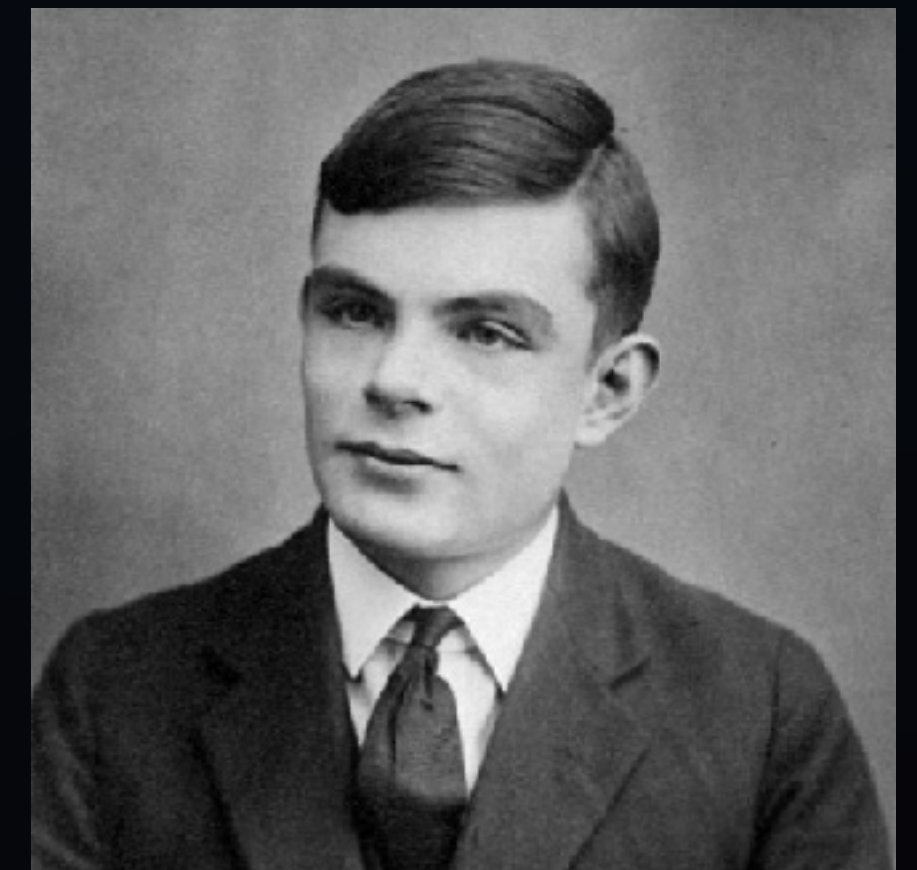
- “A symbolic analysis of relay and switching circuits” (Shannon, 1938)
 - Master’s thesis of the century
- Electronic machine could express in terms of Boolean equations with closed and open states of a circuit.
 - The programming as *a problem of formal logic* rather than of arithmetic.
- *“Shannon had injected a subject of purely academic interest into the world of practical machinery.”*

Claude Elwood Shannon
(1916–2001)



The father of artificial intelligence 🇬🇧

- Alan Turing (1912–1954)
 - *“Any explicitly stated computational task could be performed by a machine in possession of an appropriate finite set of instructions.”*
 - Church–Turing thesis* — *“Anything that can be computed algorithmically can be computed by a Turing machine.”*
- On the relationship between human thought and machine thought
 - Turing test (Turing, 1950) to distinguish the answers of humans or machines



*https://en.wikipedia.org/wiki/Church–Turing_thesis

Repercussions of the day

- Warren McCulloch and Walter Pitts (1943) were working on *neural networks*.
 - computations with a suitable finite networks of neurons
 - as a Turing machine (McCorduck 1979)
- John von Neumann (1903–1957)
 - *A stored program* housed in the computer's internal memory
 - The breakthroughs for *assemblers and compilers*, pursuing with the analogies between the brain and computing machines.
 - Worked on the Manhattan Project



Von Neumann's wartime
Los Alamos ID badge
photo (1942–1946)

Von Neumann in the 1940s

Other milestones in 60-70s

- Conversational models
- Kenneth Colby (Stanford) and Joseph Weizenbaum (MIT) wrote programs that modeled two-agent dialogues.
- In the light of Turing's famous imitation game, it was a tempting goal to pursue.

Colby's personality chatbot

- Colby's chatbot have psychiatric disorders and so the agent is given by a paranoid personality with an obsession with Mafia:

Why are you in the hospital?

I SHOULDN'T BE HERE.

Who brought you here?

THE POLICE.

What trouble did you have with the police?

COPS DON'T DO THEIR JOB.

Weizenbaum's *ELIZA*, the therapist

- Weizenbaum's chatbot was imitating a *Rogerian therapist*, who is performing non-directive but supportive therapy sessions.

Men are all alike.

IN WHAT WAY?

They're always bugging us about something or other.

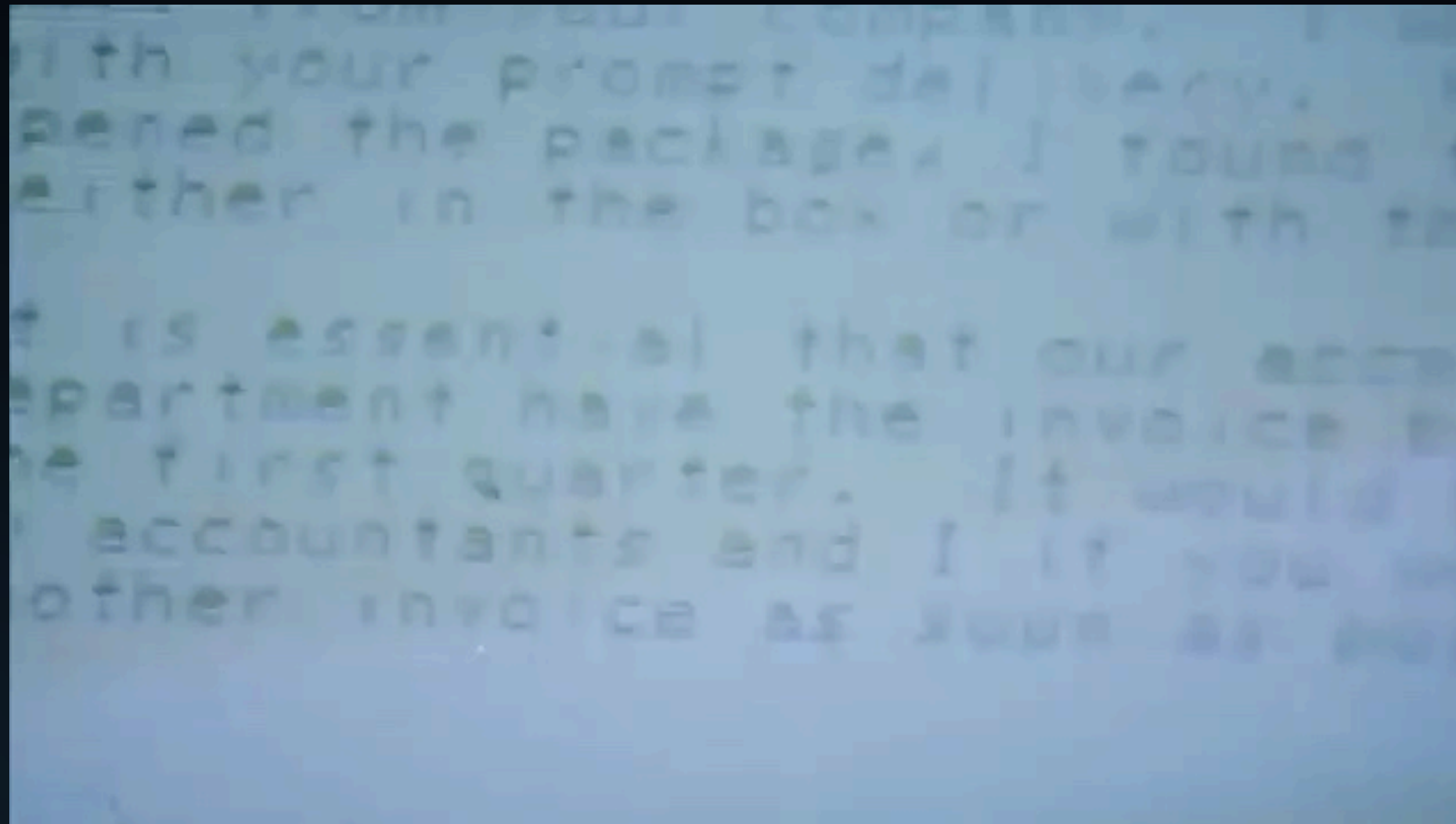
CAN YOU THINK OF A SPECIFIC EXAMPLE?

Well, my boyfriend made me come here.

YOUR BOYFRIEND MADE YOU COME HERE.

Weizenbaum's *ELIZA*, the therapist

- Before ChatGPT, there was ELIZA: the computer therapist chatbot written by Joseph Weizenbaum at MIT's AI Lab in 1966.

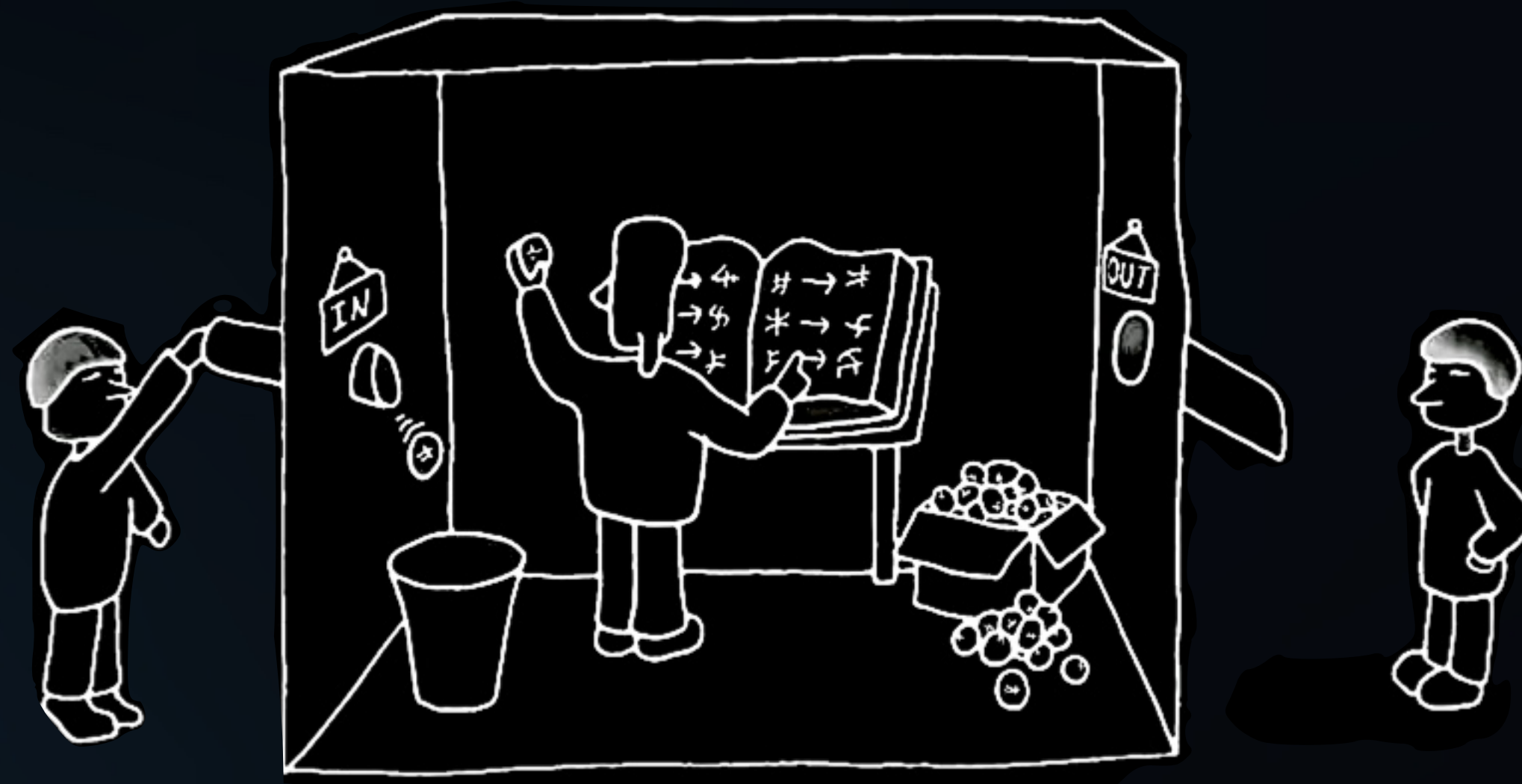


Unending debates

- ELIZA depended on *pattern-matching* techniques and sometimes responds with content-free rules, which may be aligned with the *Rogsonian* therapist.
- Colby and Weizenbaum initially collaborated, but it diverged in the attitudes toward AI. Weizenbaum posed rather to be practical, aside from the field of AI.
- Colby was a *true believer* who thought that AIs would play an important role in the treatment research of mental illness.
- The debates on the roles and values of AI are still incurring.

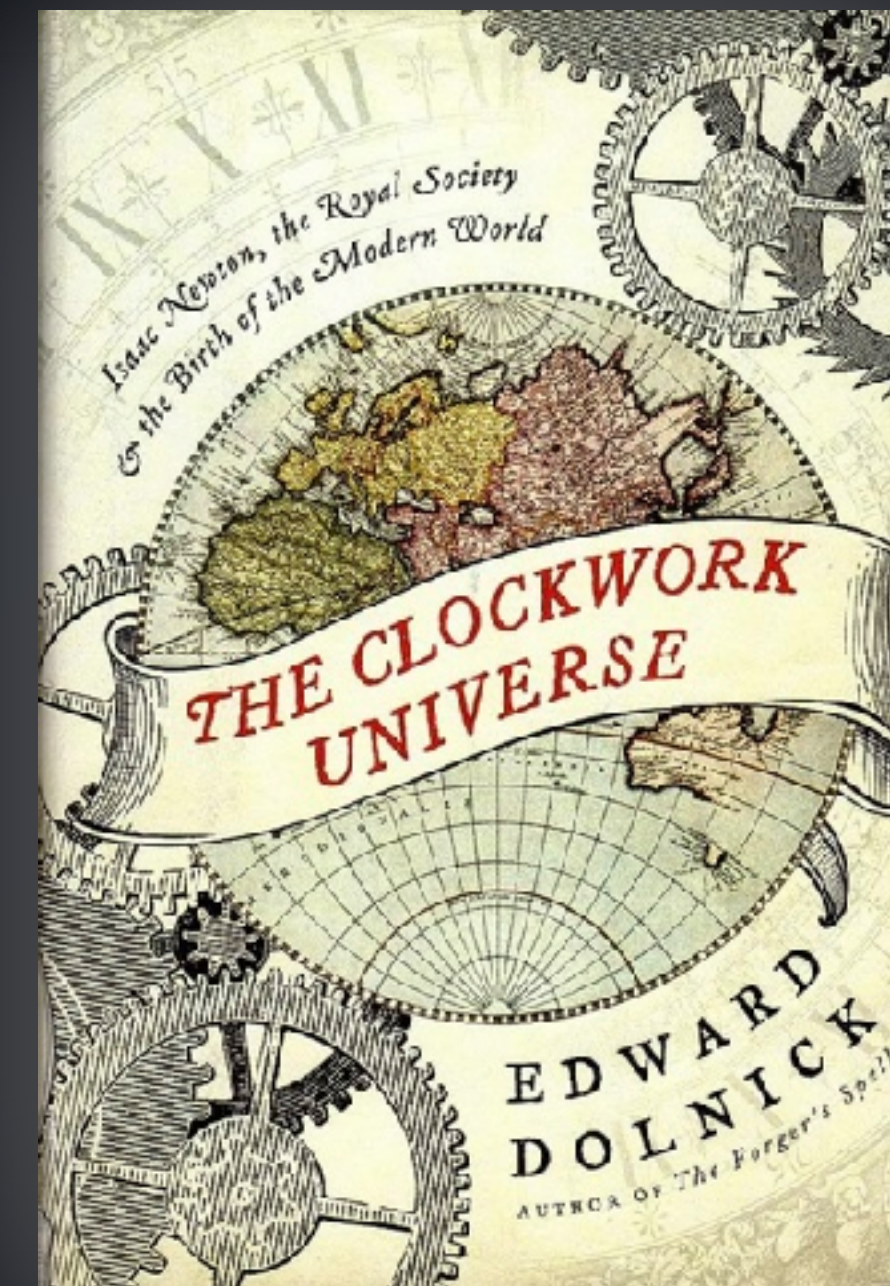
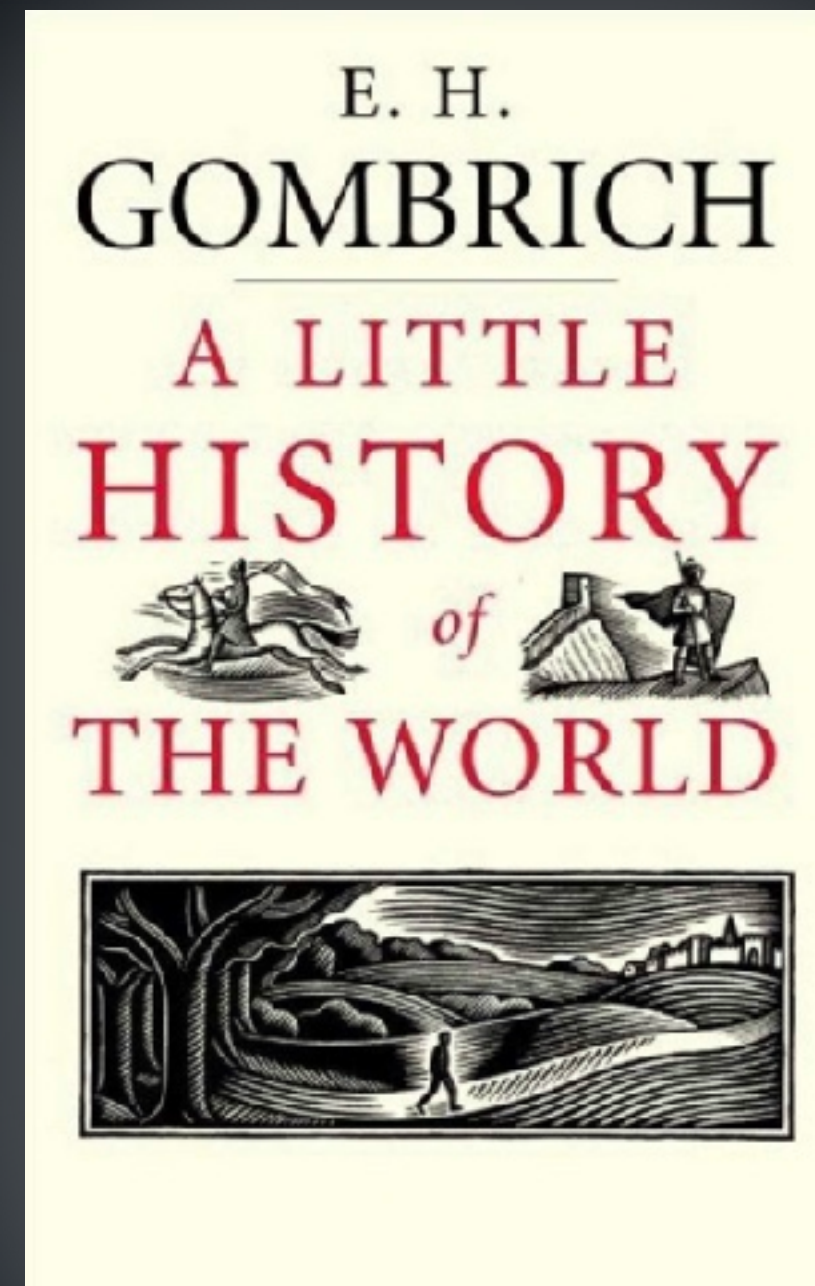
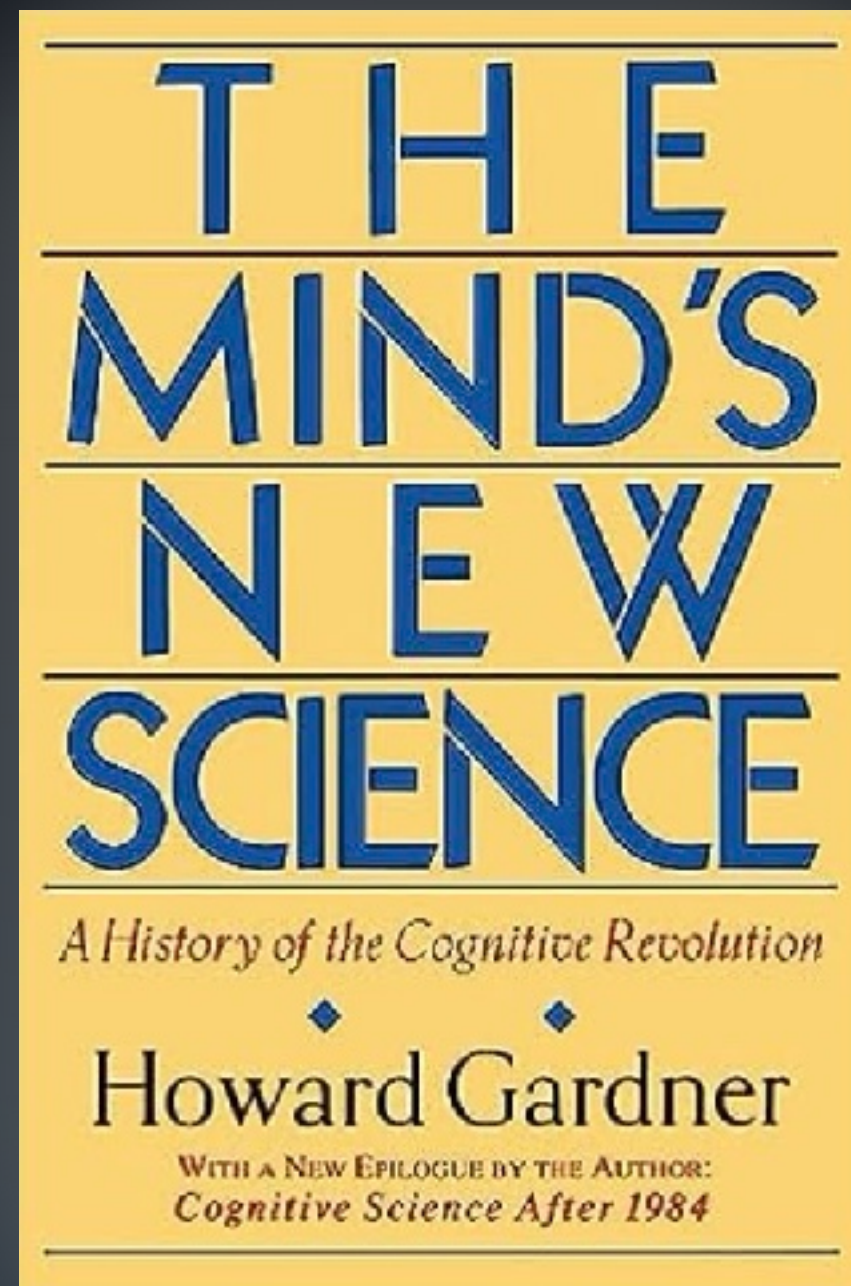
Searle's Chinese room

- A *thought experiment* to counter to the Turing test and against the *functionalism* and *computationalism*.
- How do we define “understand” and “intentionality” of agents?



Inspiring books

If your curiosity isn't quite satisfied yet...

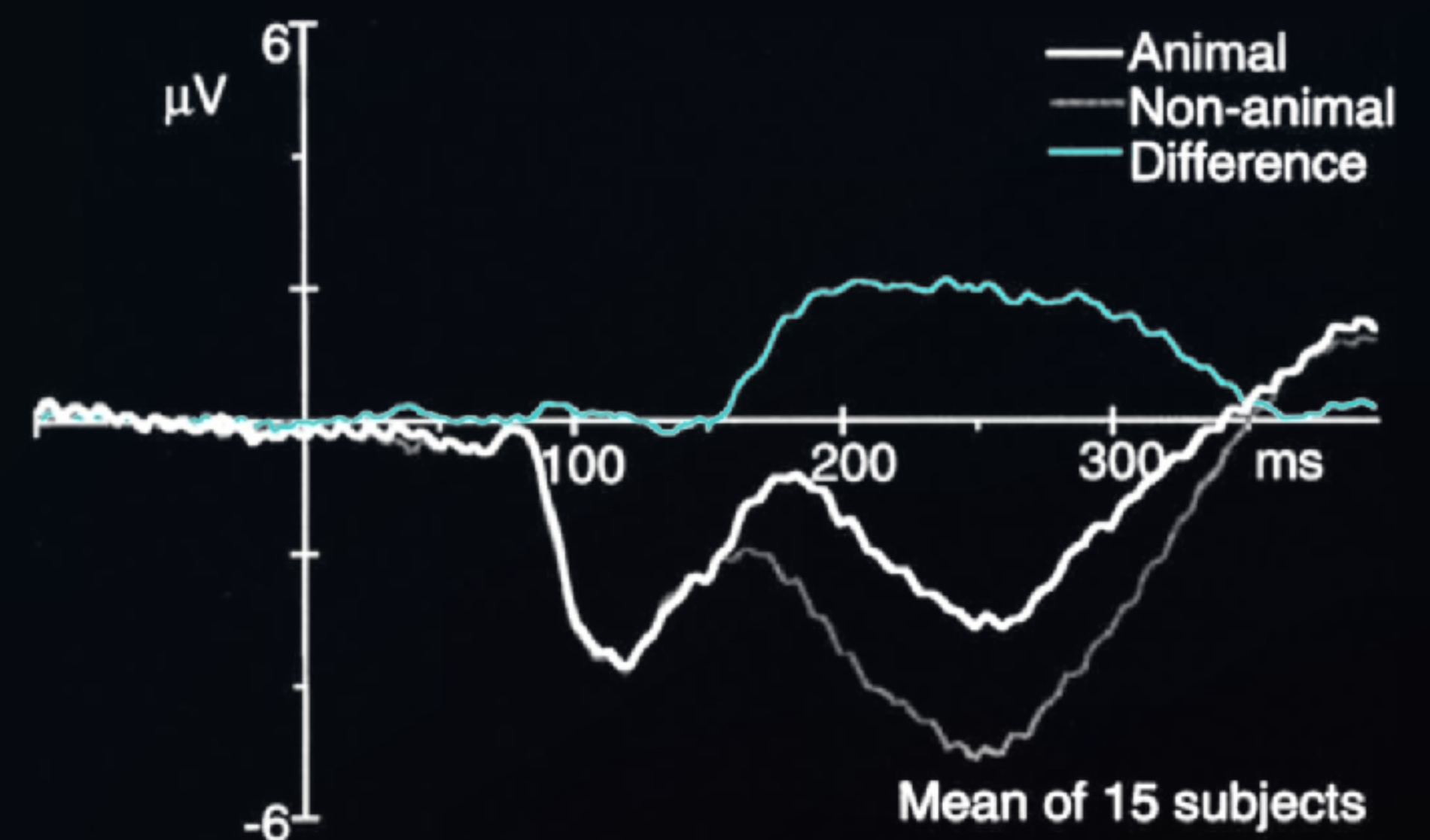


Computer Vision

This section is inspired by Fei-Fei's invited talk at NeurIPS 2024.

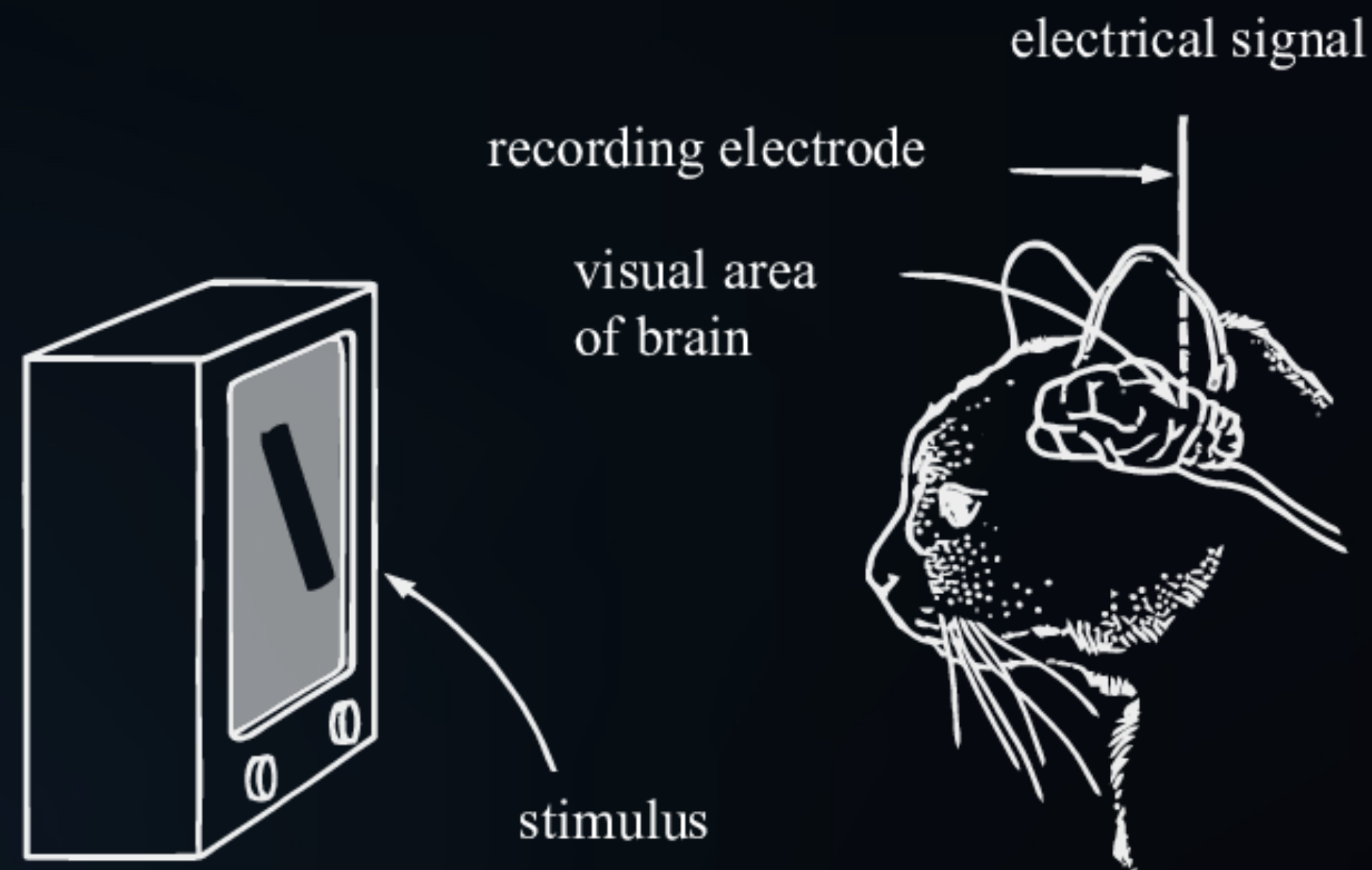
Speed of objects categorization

- Object categorization occurs as early as 150 ms.
 - By observing the event-related potentials (ERPs) with a go/no-go categorization task.
- The complexity of the visual recognition pipeline may be:
 - Retinal encoding (0-20 ms)
 - Optic nerve transmission (20-30 ms)
 - Early visual cortex processing (30-70 ms)
 - High-level visual processing (70-120 ms)
 - Categorization response (120-150 ms)



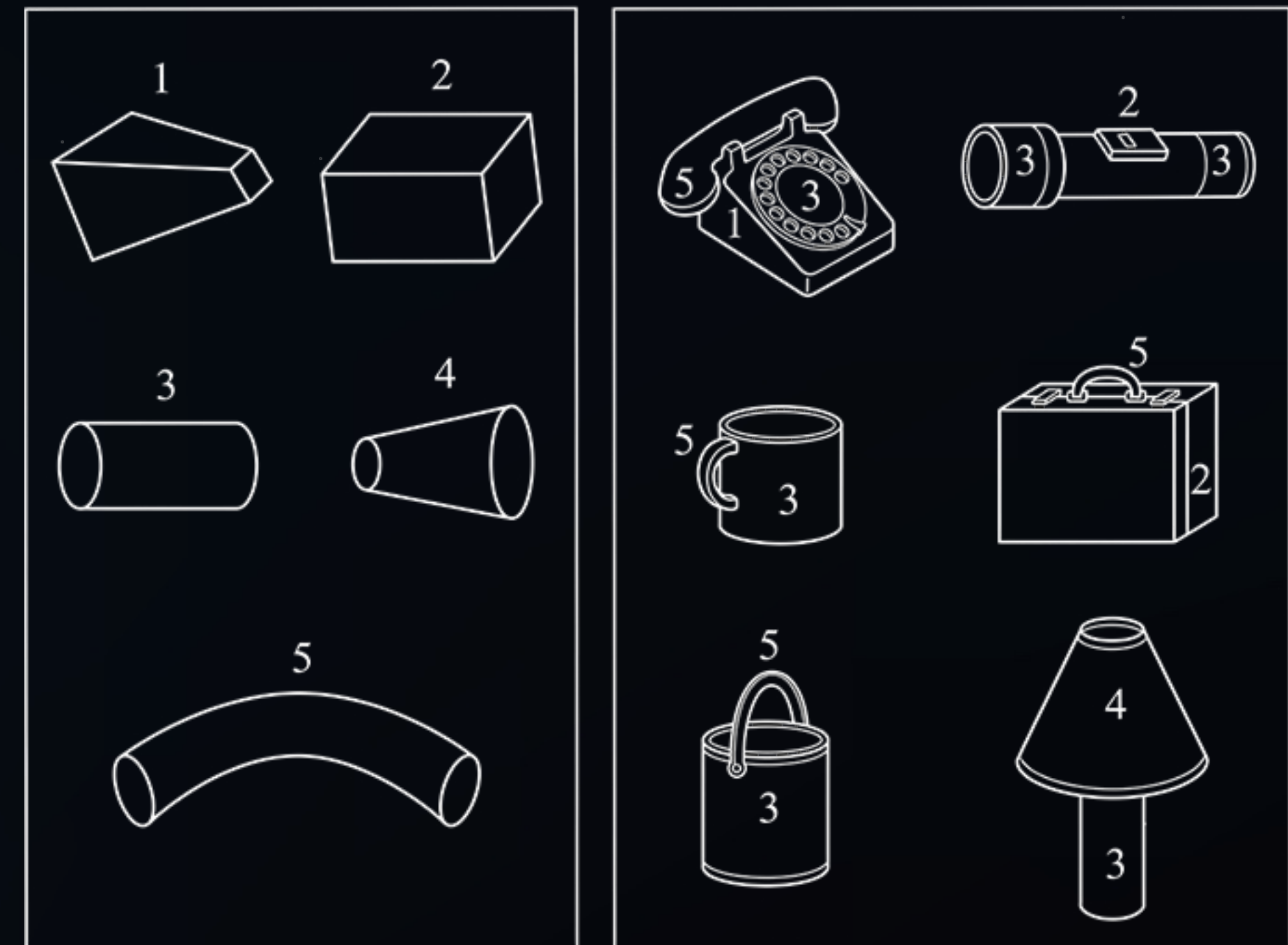
Individual cells respond to certain shapes

- In 1959, Hubel and Wiesel discovered specialized cells in the visual cortex.
- Some cells respond only to horizontal lines, others only to vertical lines.
- Their groundbreaking study earned them the *Nobel Prize in Physiology or Medicine in 1981!*



Early models for object recognition

- In psychology, *Geons* refer to simple 2D or 3D forms.
- Recognition-by-components (RBC) theory posits that the geons serve as the primary components of objects (Biederman, 1987).
- ACRONYM (Brooks et al., IJCAI 1979)
- Representing shape for recognition (Marr & Nishihara, 1978)



Early models for object recognition (Cont'd)

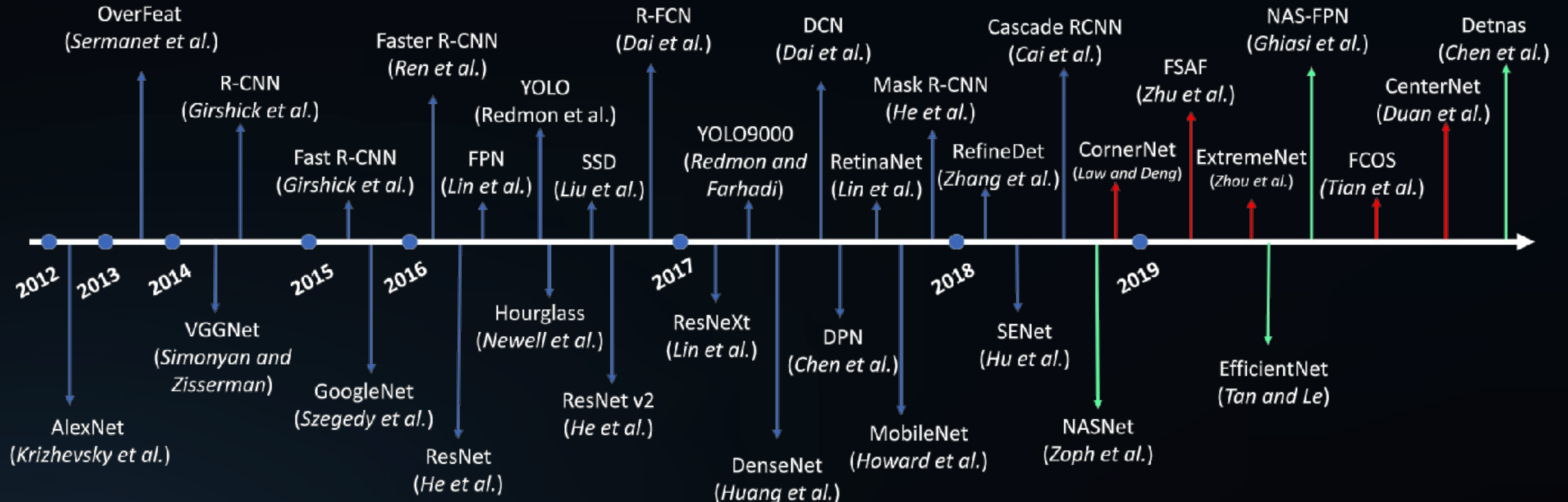
- Bag-of-words (Felzenszwalb et al., 2000; Fergus et al., 2003; Fei-Fei et al., 2003)
- Boosting (Chen et al., 2006; Zhu et al., 2007)
- Non-parameteric Bayes (Viola & Jones, 2001; Torralba et al., 2004)



Object classification datasets

- SUN 131 K (Xiao et al., 2010)
- LabelMe 37 K (Russell et al., 2007)
- PASCAL VOC 30 K (Everingham et al., 2006-2012)
- Caltech101 9K (Fei-Fei, Fergus, & Perona, 2003)
- ImageNet 15 M images w/ 22k categories (Deng et al., CVPR 2009)

Challenges drive progress



Multimodal deep learning

McGurk effect

- McGurk and MacDonald ([Nature, 1976](#)) showed a multi-sensory illusion highlighting the quality difference of bimodal stimuli. *"Hearing Lips and Seeing Voices"*
- A vision of /ga/ with a voice of /ba/ *is perceived as* /da/ by most subjects.
- Vision "provides information on the place of articulation and muscle movements, which can help to *disambiguate* between speech with similar acoustics (e.g., the unvoiced consonants /p/ and /k/)." ([Summerfield, 1992](#))
- *Ref. the ventriloquism effect*



Binding problem

- “The binding problem is from how mammals (particularly higher primates) generate a **unified perception** of their surroundings from electromagnetic waves, chemical interactions, and pressure fluctuations that forms the physical basis of the world around us. (rephrased)”
- Classical conditioning, Pavlov’s dog:

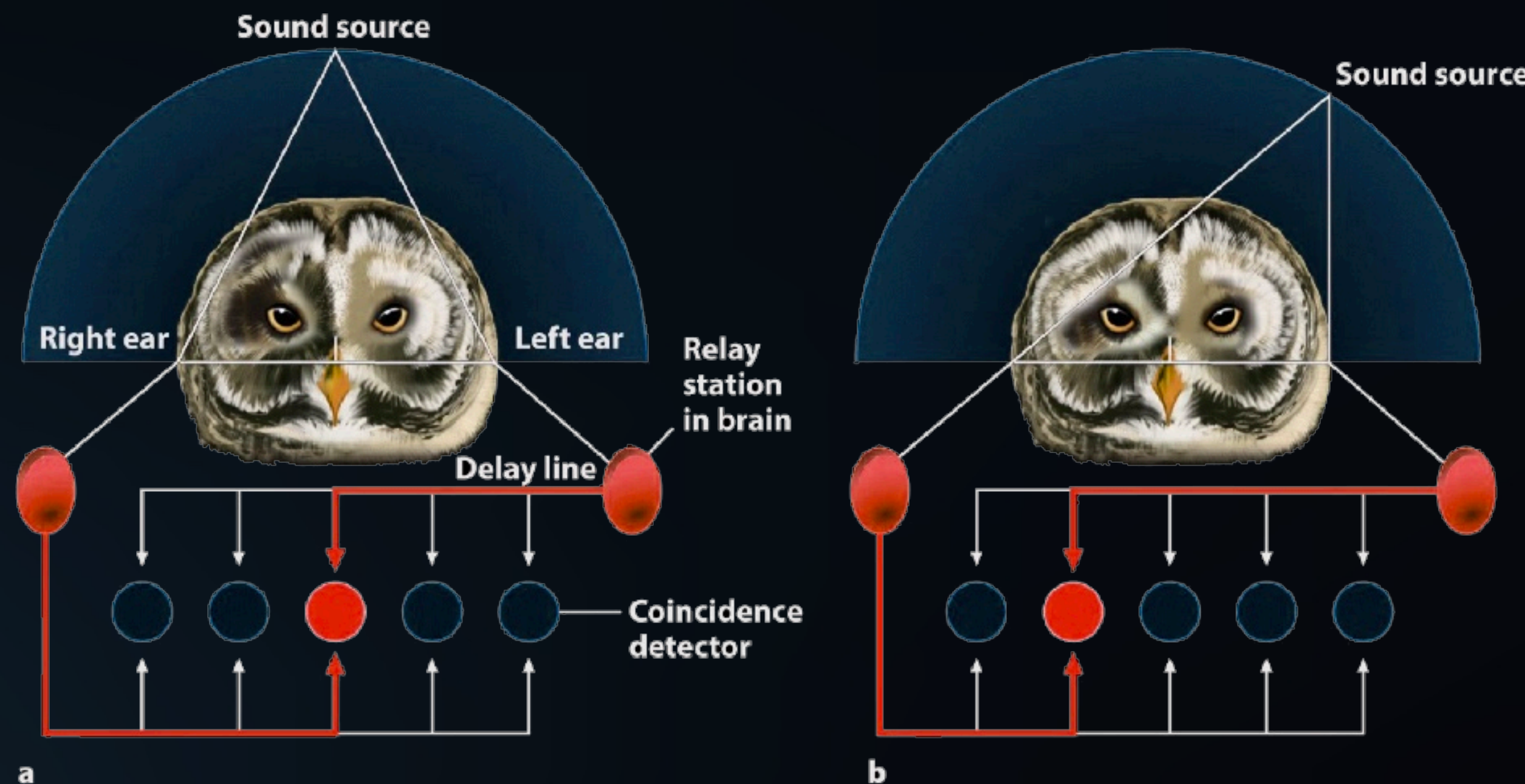


Timing is everything

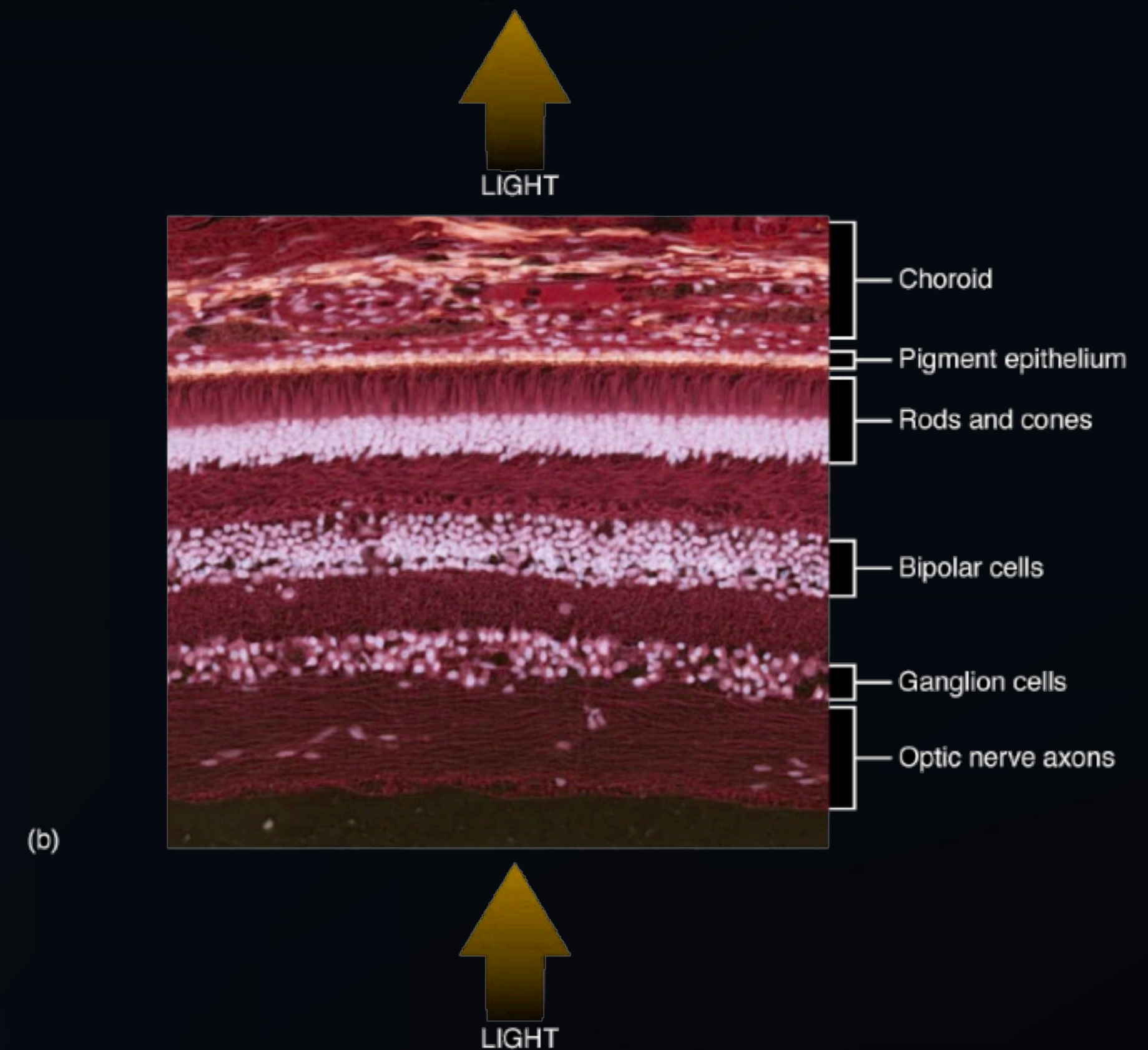
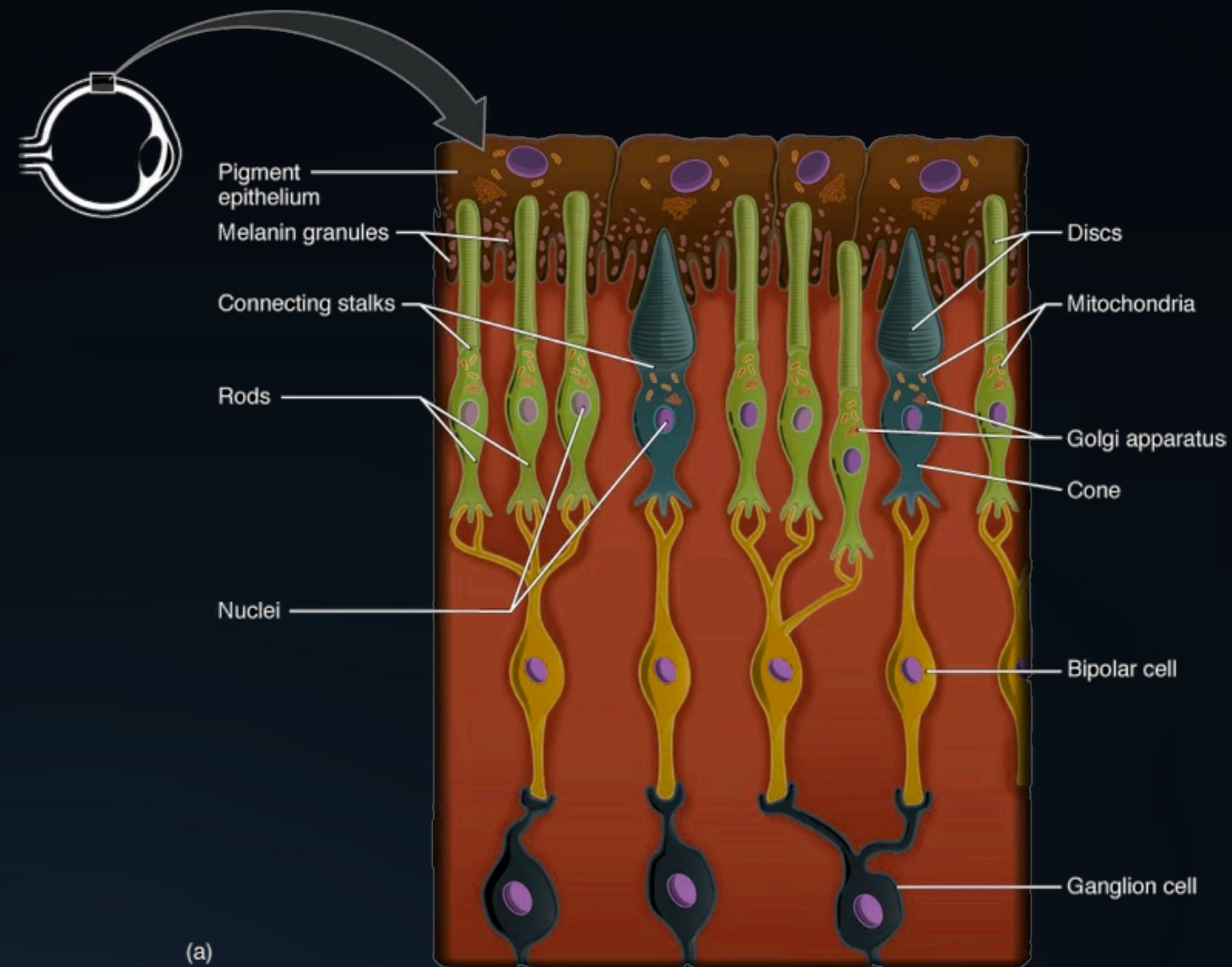
- Timing is critical to the binding problem ([Vroomen & Keetels, 2010](#)).
- In physiology, the binding problem is generally referred to temporal synchrony:
 - the coincidence of neural firing,
 - frequent patterns in large scale neural activities.

Sound localization in owls

- A example is the *coincidence detector* of barn owls (Carr & Konishi, 1990).
- Using inter-aural time and the difference in sound's intensity at two ears, the asymmetric cues are exploited to localize the source of sound in dark.



Photoreceptor cell



Representation binding

- In deep learning, learning joint representation to solve the binding problem in multimodal learning tasks, e.g., visual question answering (will be discussed).
- *Mid-level* embedding for each modality is crucial (Ngiam et al., 2011).
- Large models for each modality is for state-of-the-art performance (Chen et al., 2022).
- Early multimodal learning methods may include:
 - Bilinear models, cross-attention models, and
 - Self-supervised learning maximizing mutual information.

Large Language Models

ChatGPT (2022)

ChatGPT



Examples

"Explain quantum computing in simple terms" →

"Got any creative ideas for a 10 year old's birthday?" →

"How do I make an HTTP request in Javascript?" →



Capabilities

Remembers what user said earlier in the conversation

Allows user to provide follow-up corrections

Trained to decline inappropriate requests



Limitations

May occasionally generate incorrect information

May occasionally produce harmful instructions or biased content

Limited knowledge of world and events after 2021



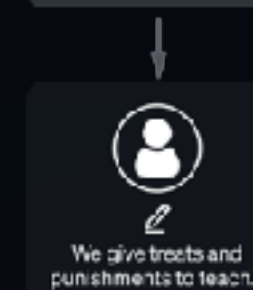
ChatGPT Dec 15 Version. Free Research Preview. Our goal is to make AI systems more natural and safe to interact with. Your feedback will help us improve.

Step 1
Collect demonstration data and train a supervised policy.

A prompt is sample from our prompt dataset.



A labeler demonstrates the desired output behavior.

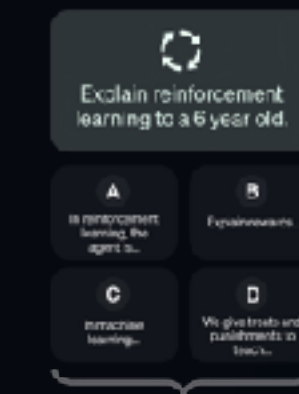


This data is used to fine-tune GPT-3.5 with supervised learning.

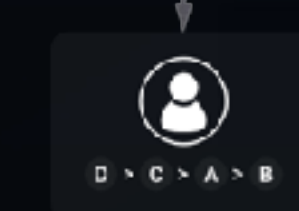


Step 2
Collect comparison data and train a reward model.

A prompt and several model outputs are sampled.



A labeler ranks the outputs from best to worst.



This data is used to train our reward model.

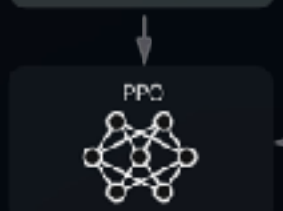


Step 3
Optimize a policy against the reward model using the PPO reinforcement learning algorithm.

A new prompt is sampled from the dataset.



The PPO model is initialized from the supervised policy.



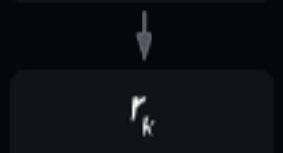
The policy generates an output.



The reward model calculates a reward for the output.



The reward is used to update the policy using PPO.

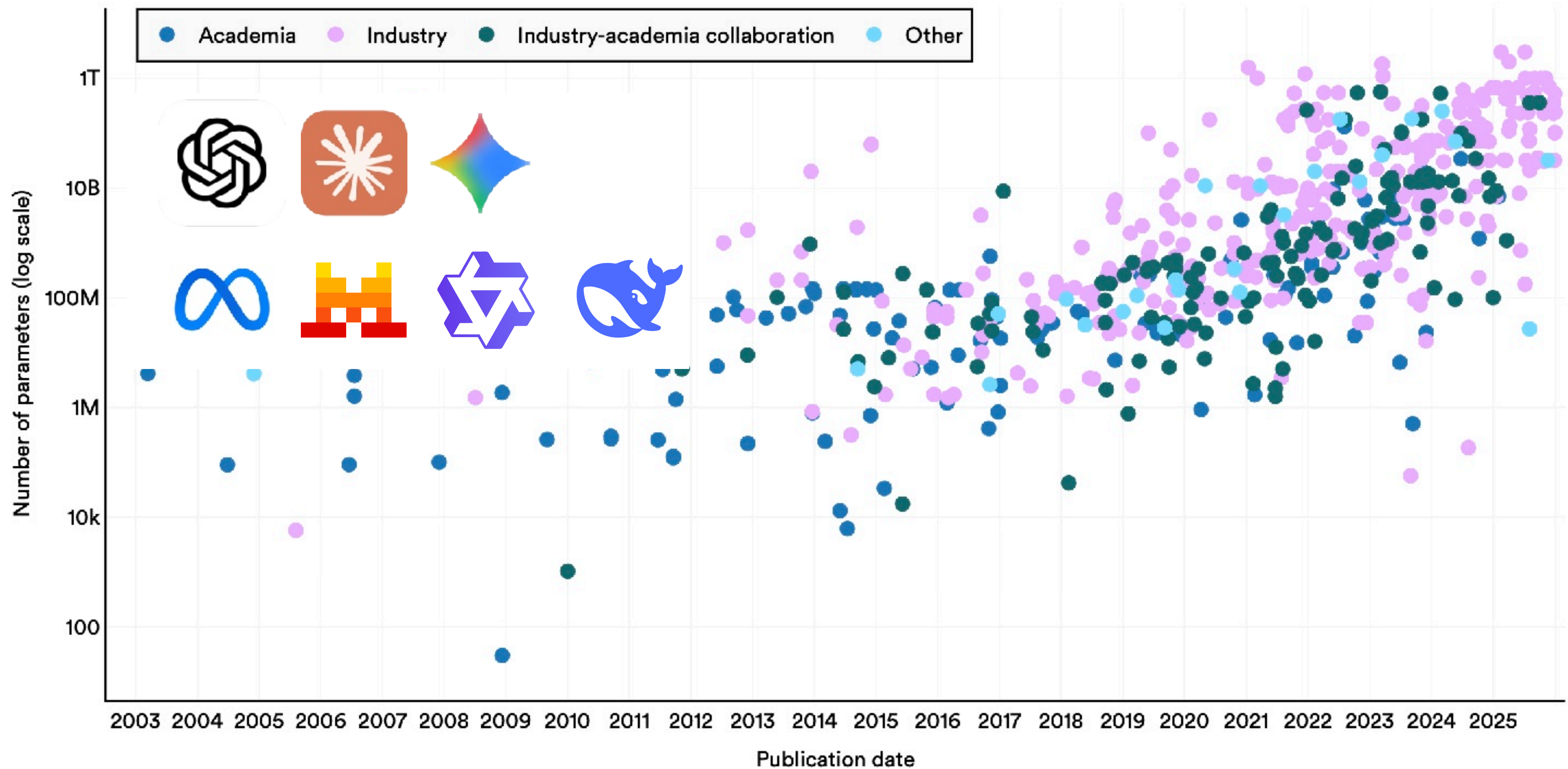


GPT-3.5 (RLHF)

Large language models

Number of parameters of notable AI models by sector, 2003–25

Source: Epoch AI, 2026 | Chart: 2026 AI Index report



AAAI'26 adopts AI-powered review system

- “Enhancing scientific Review, not replacing human expertise”
 - Supplementary first-stage reviews and discussion summary assistance
- “Preserving human decision-making and scientific integrity”
- “cutting-edge methods with rigorous safeguards”



Multimodal foundation models

GPT-4 visual input example, Extreme Ironing:

User What is unusual about this image?



Source: <https://www.barnorama.com/wp-content/uploads/2016/12/03-Confusing-Pictures.jpg>

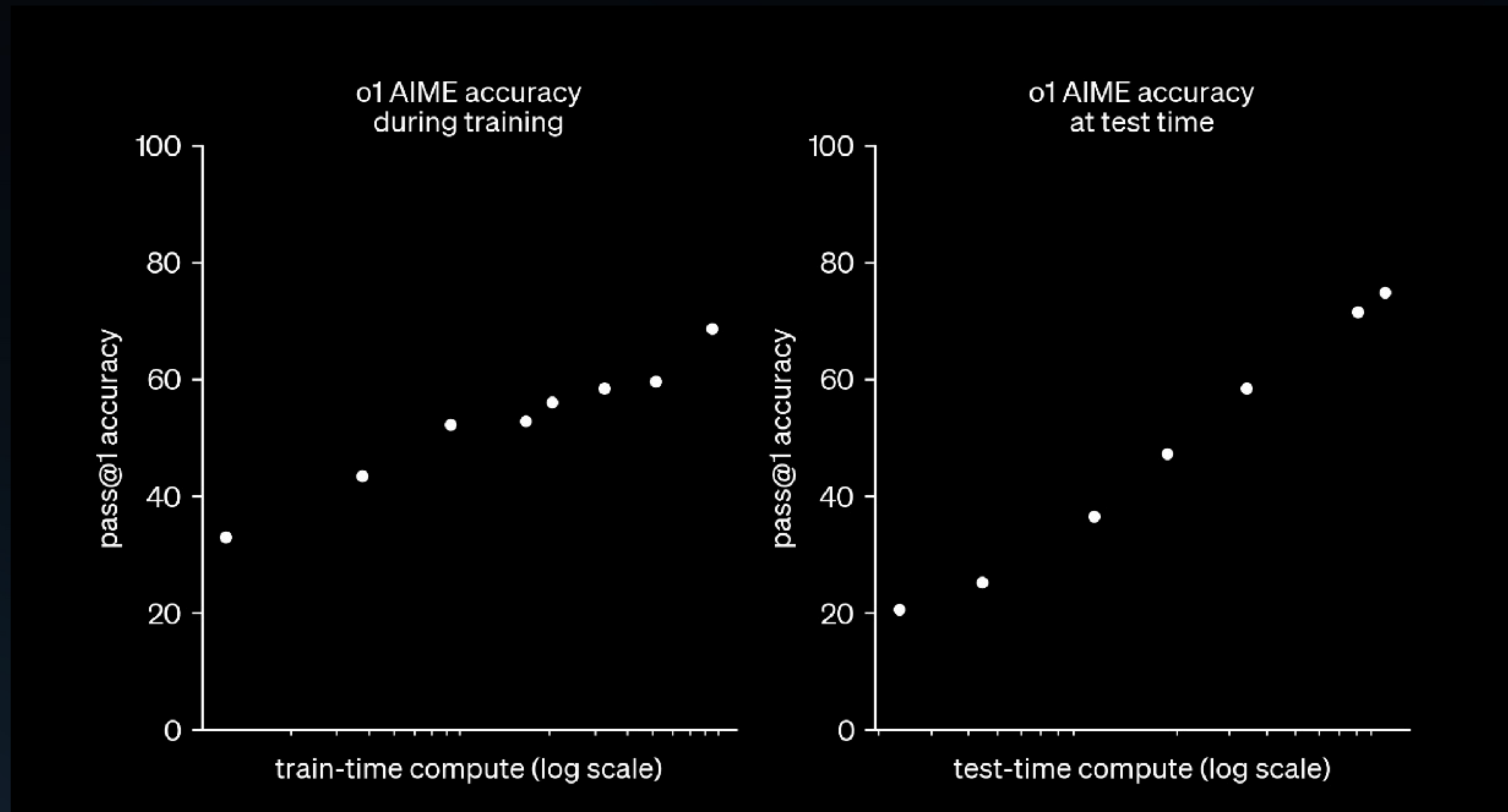
GPT-4 The unusual thing about this image is that a man is ironing clothes on an ironing board attached to the roof of a moving taxi.

GPT-4V (2023)



GPT-4o (2024)

Test-time scaling (*Thinking*)



ChatGPT o1

Thinking with images

ChatGPT o3



I took this picture earlier today. Can you find the name of the biggest ship you can see, and where it will dock next?

Identifying ship and location

- The user provided a photo and asked about the biggest ship visible, its name, and where it will dock next. Since the photo shows multiple cargo ships near a waterfront, the task is tricky. The name on the ships might be too small or far away, and without a specialized tool for ship recognition, I may need to rely on AIS tracking, since the photo could be from the San Francisco Bay Area given the building resembling Pier 15, near the Exploratorium.
- The building in the photo looks like it could be the SF James R. Herman Cruise Terminal at Pier 27, with a curved white

Message ChatGPT

Search

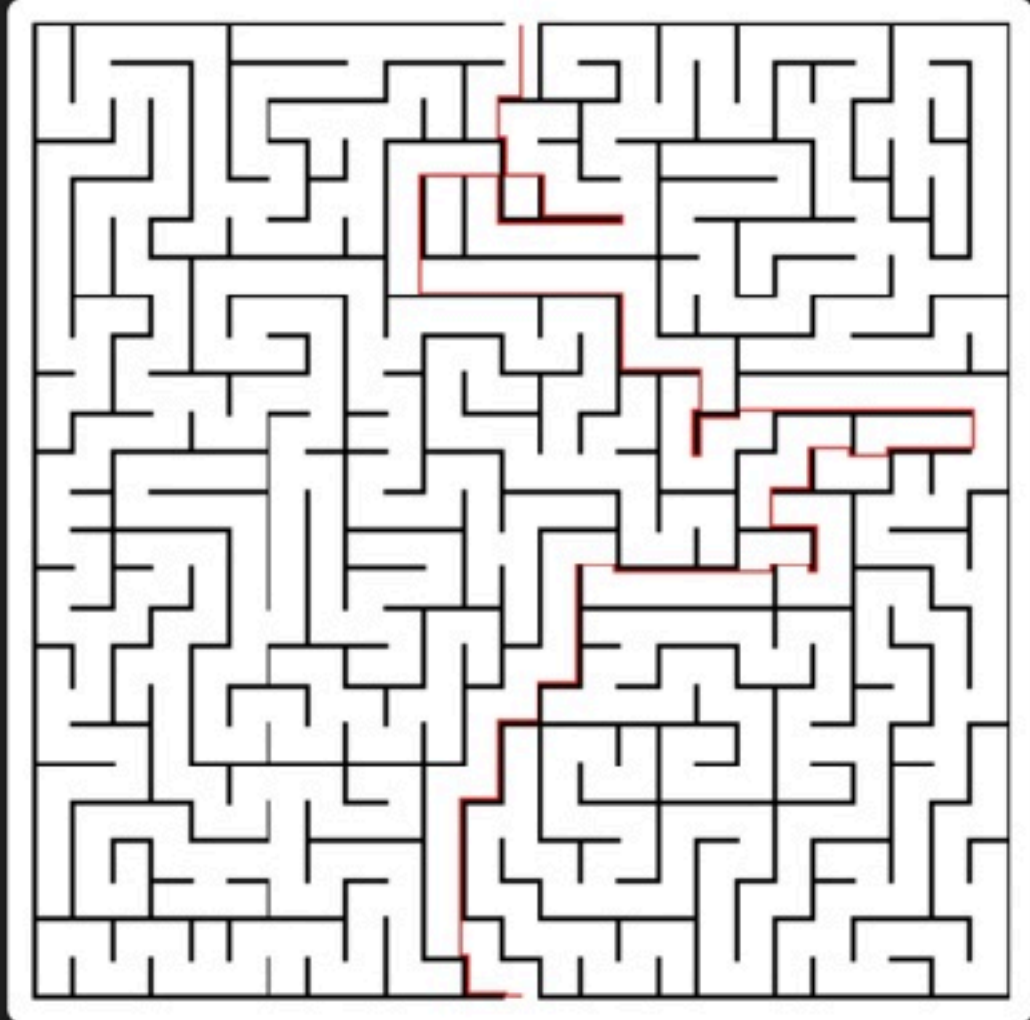
Deep research

ChatGPT can make mistakes. Check important info.

solve this maze and plot a path with red-colored line

1분 44초 동안 미로 해결 및 경로에 대해 이성 사용함

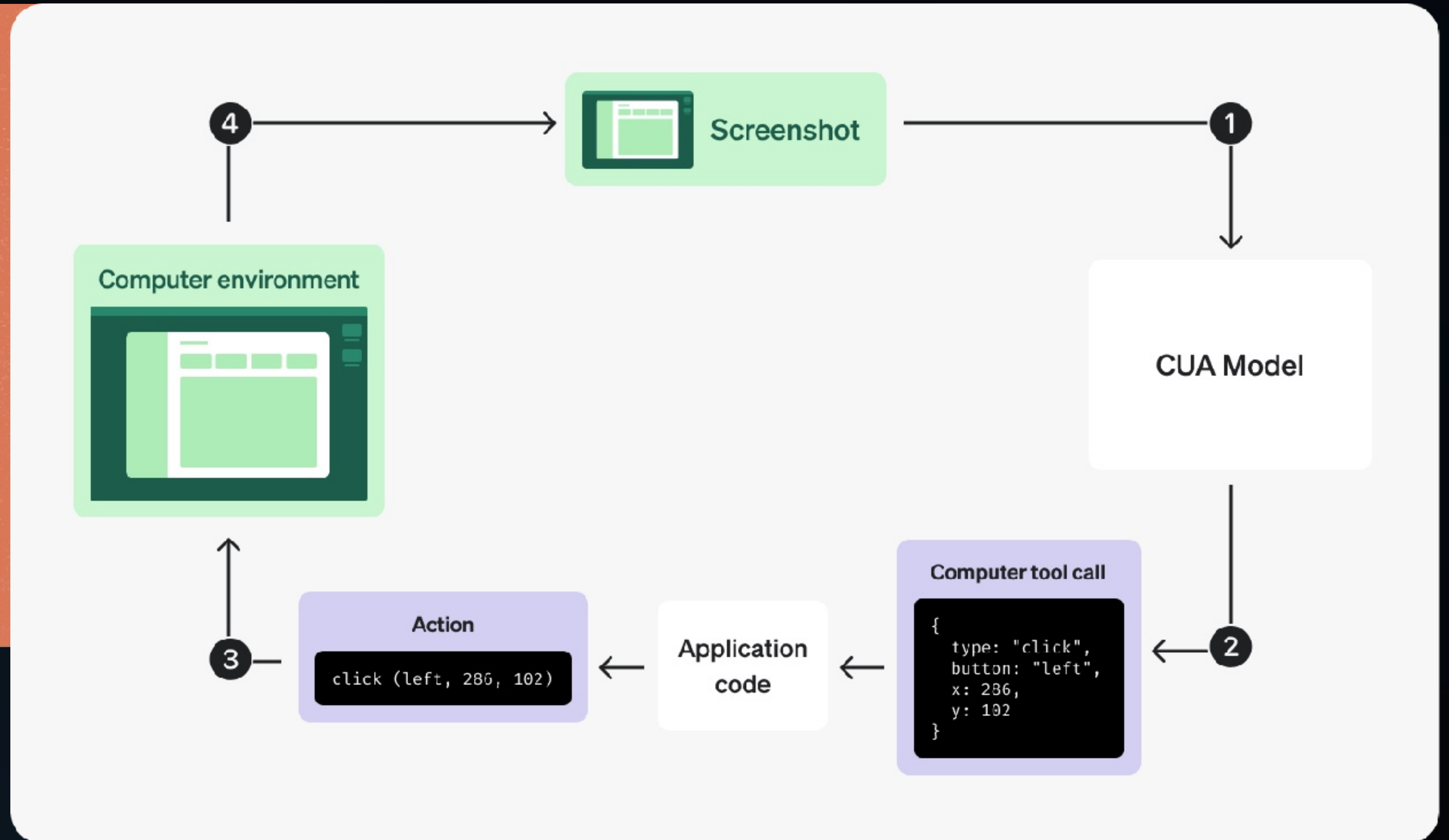
Matplotlib Chart



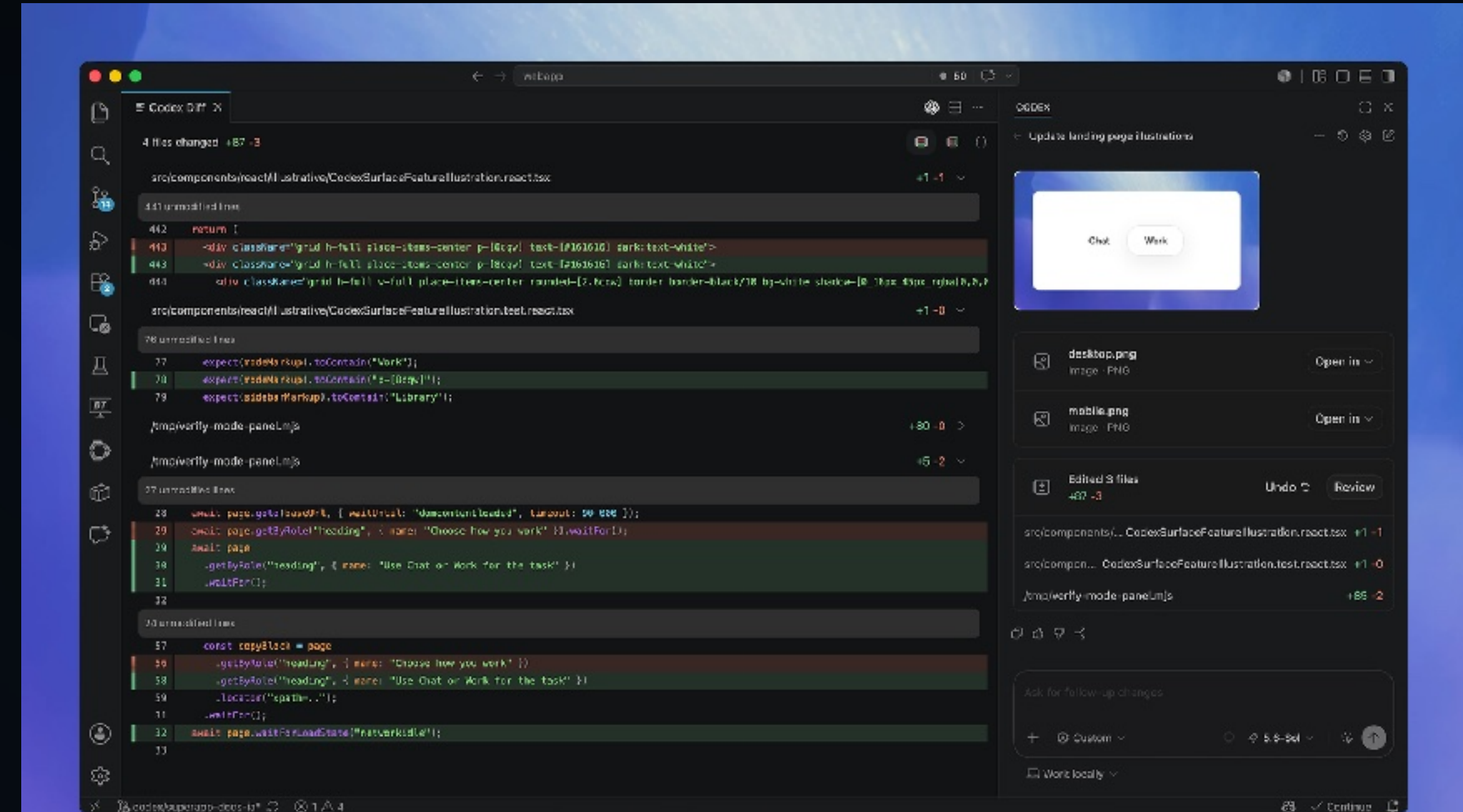
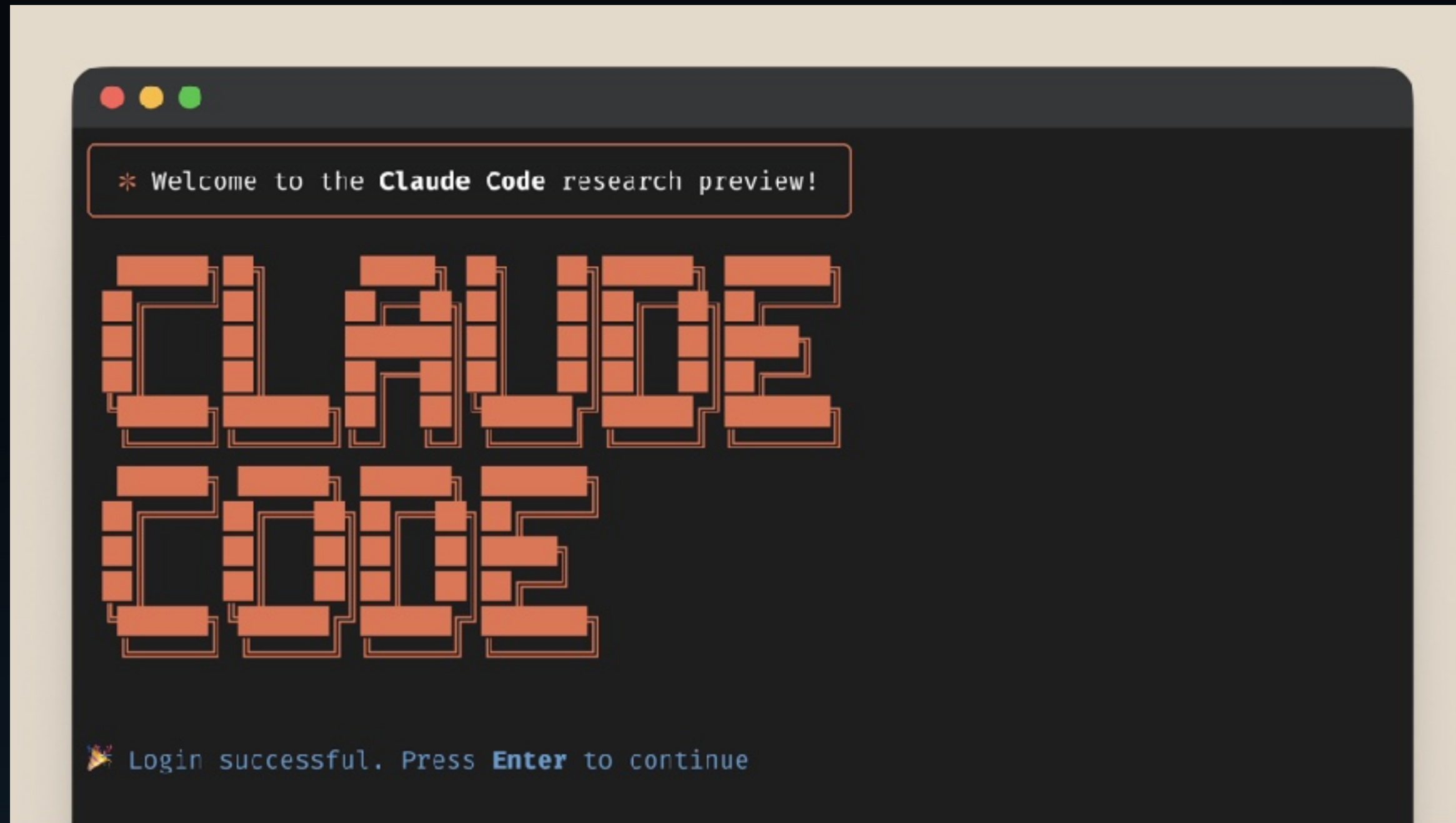
Here's one possible solution path for your maze, traced in red.

ChatGPT o3 (thinking with images)

Computer use agents



Coding agents

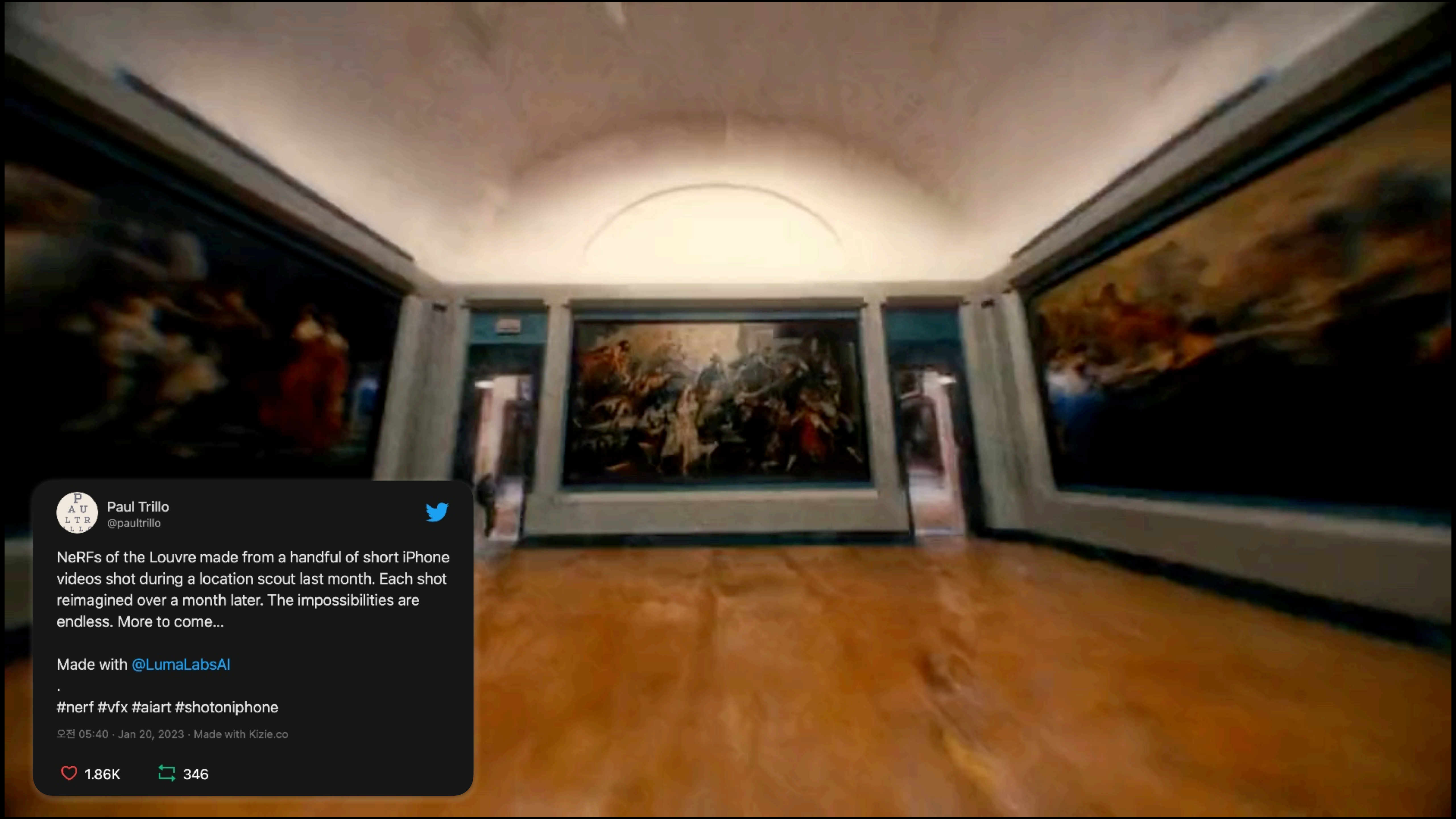


Multimodal Generation



"An A.I.-Generated Picture Won an Art Prize. Artists Aren't Happy."

By Kevin Roose, The New York Times (Sept. 2, 2022)



Paul Trillo
@paultrillo



NeRFs of the Louvre made from a handful of short iPhone videos shot during a location scout last month. Each shot reimagined over a month later. The impossibilities are endless. More to come...

Made with [@LumaLabsAI](#)

#nerf #vfx #aiart #shotoniphone

오전 05:40 · Jan 20, 2023 · Made with Kizie.co

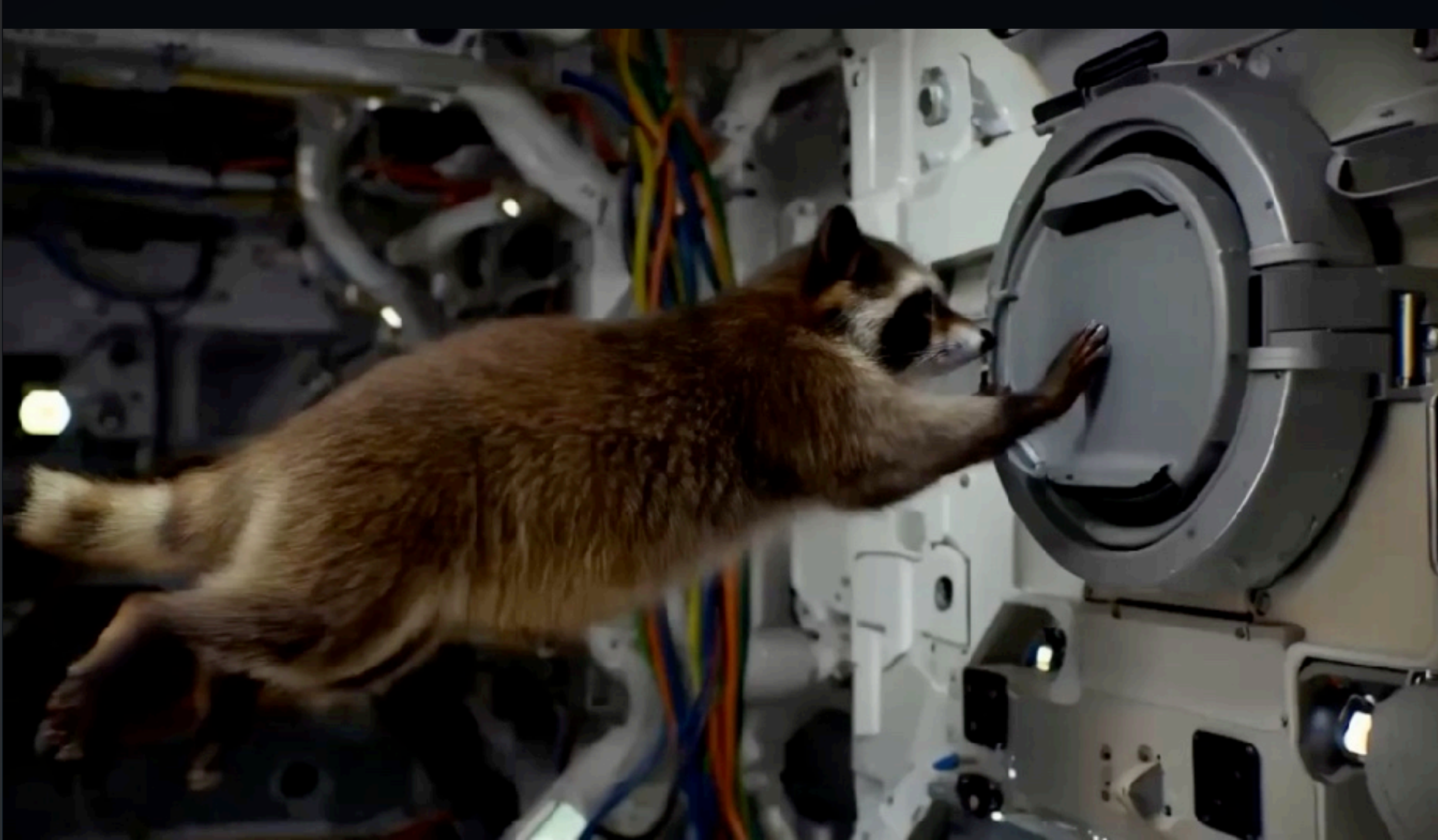


1.86K



346

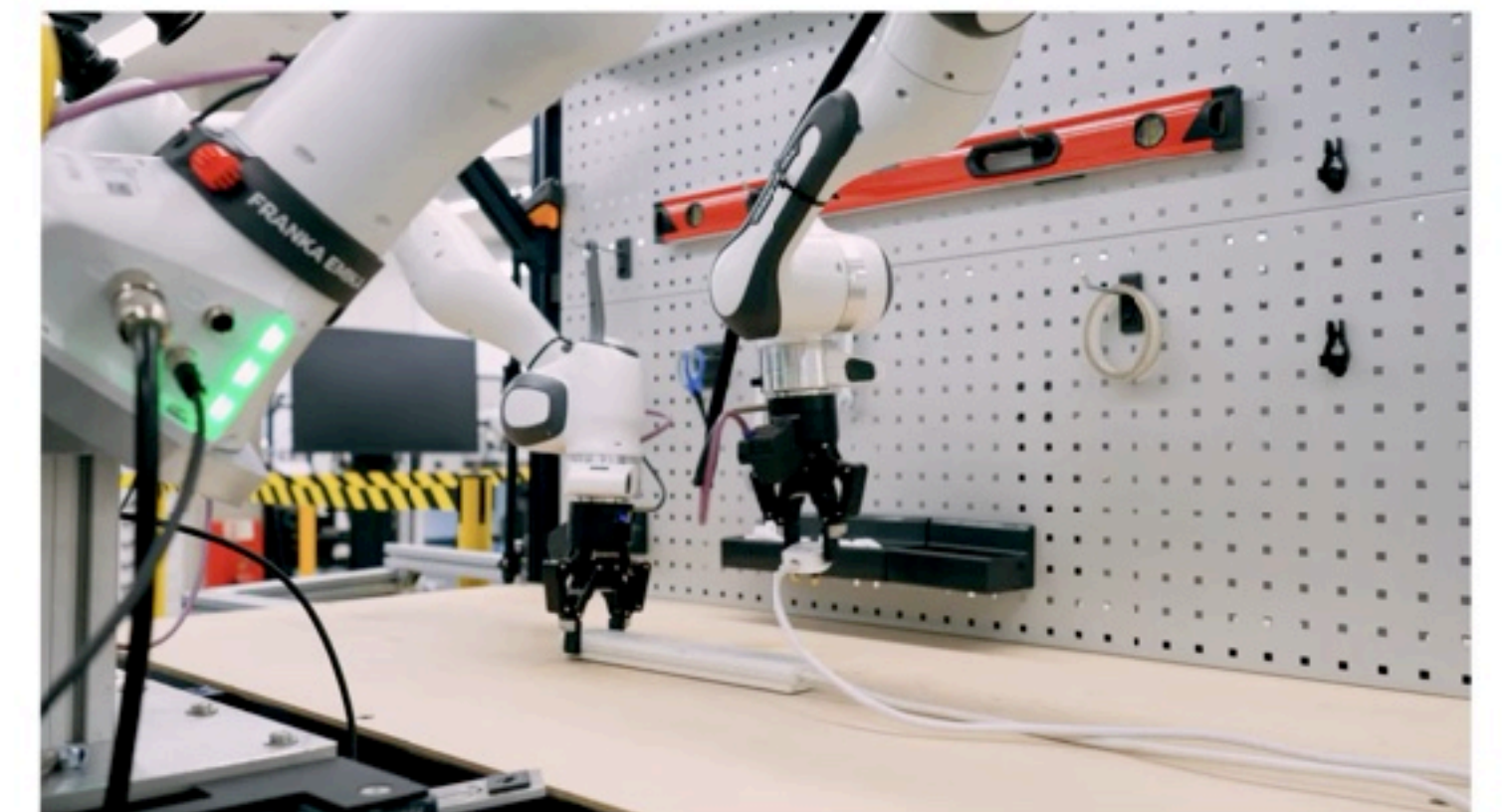
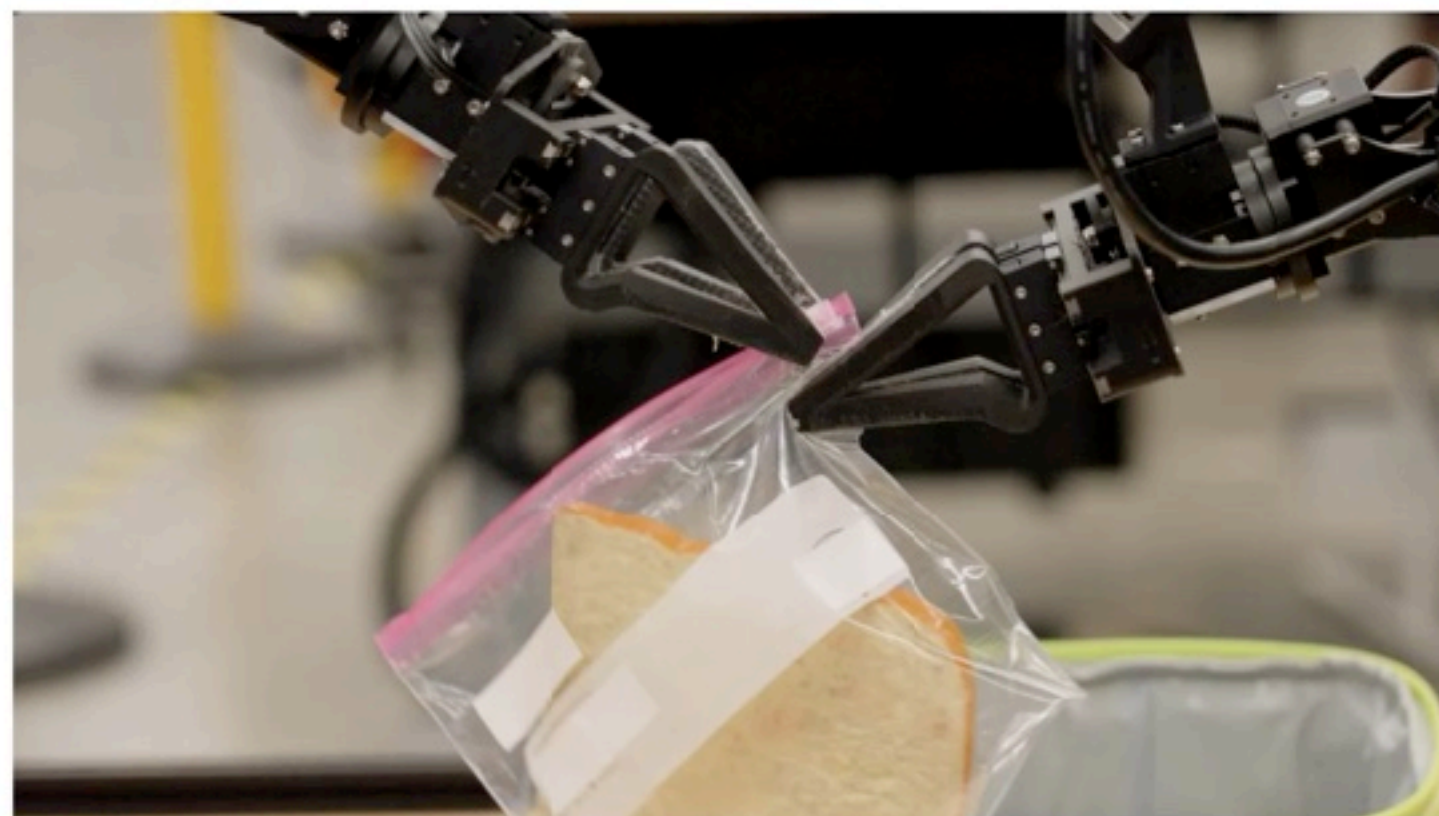
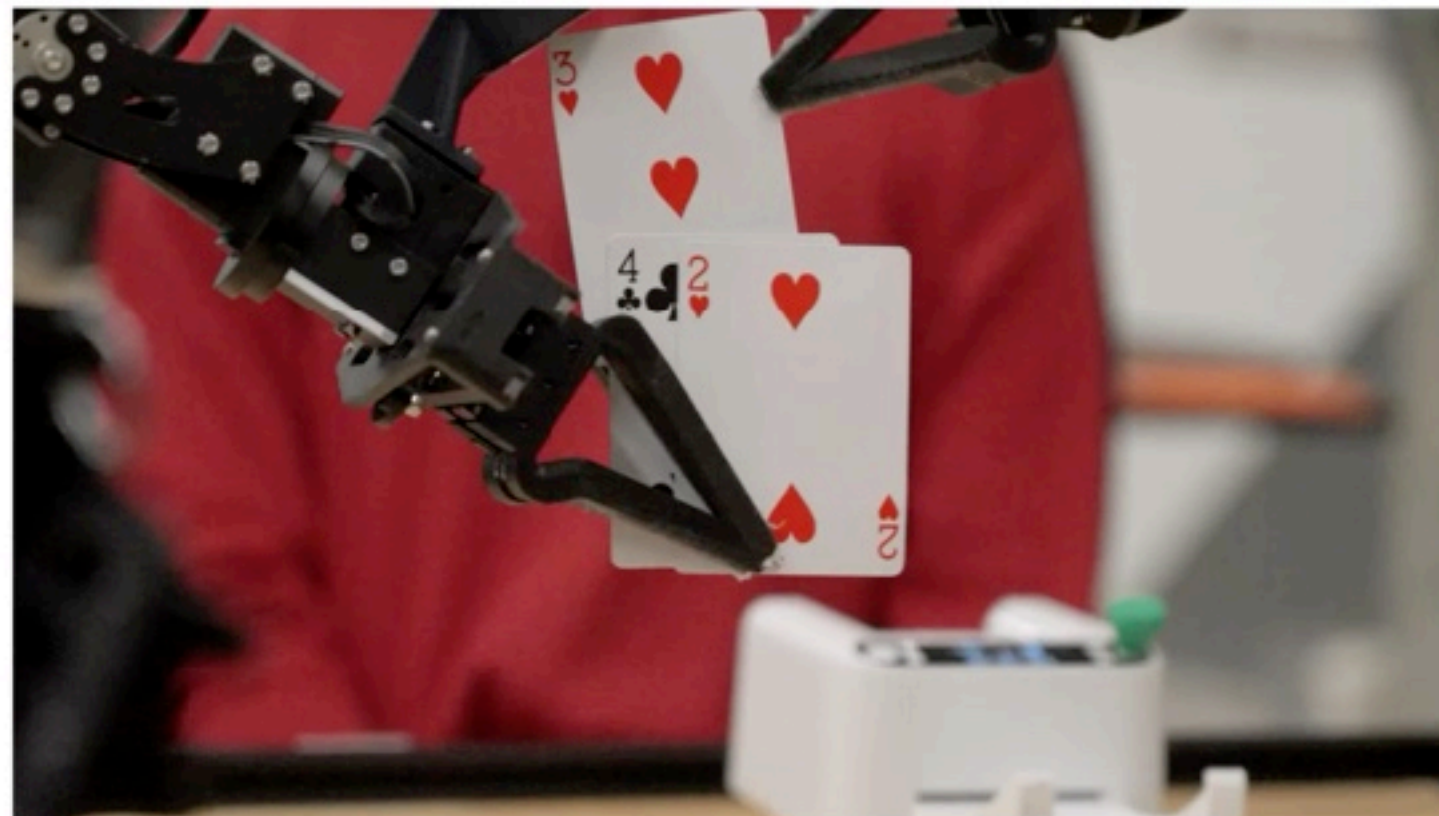
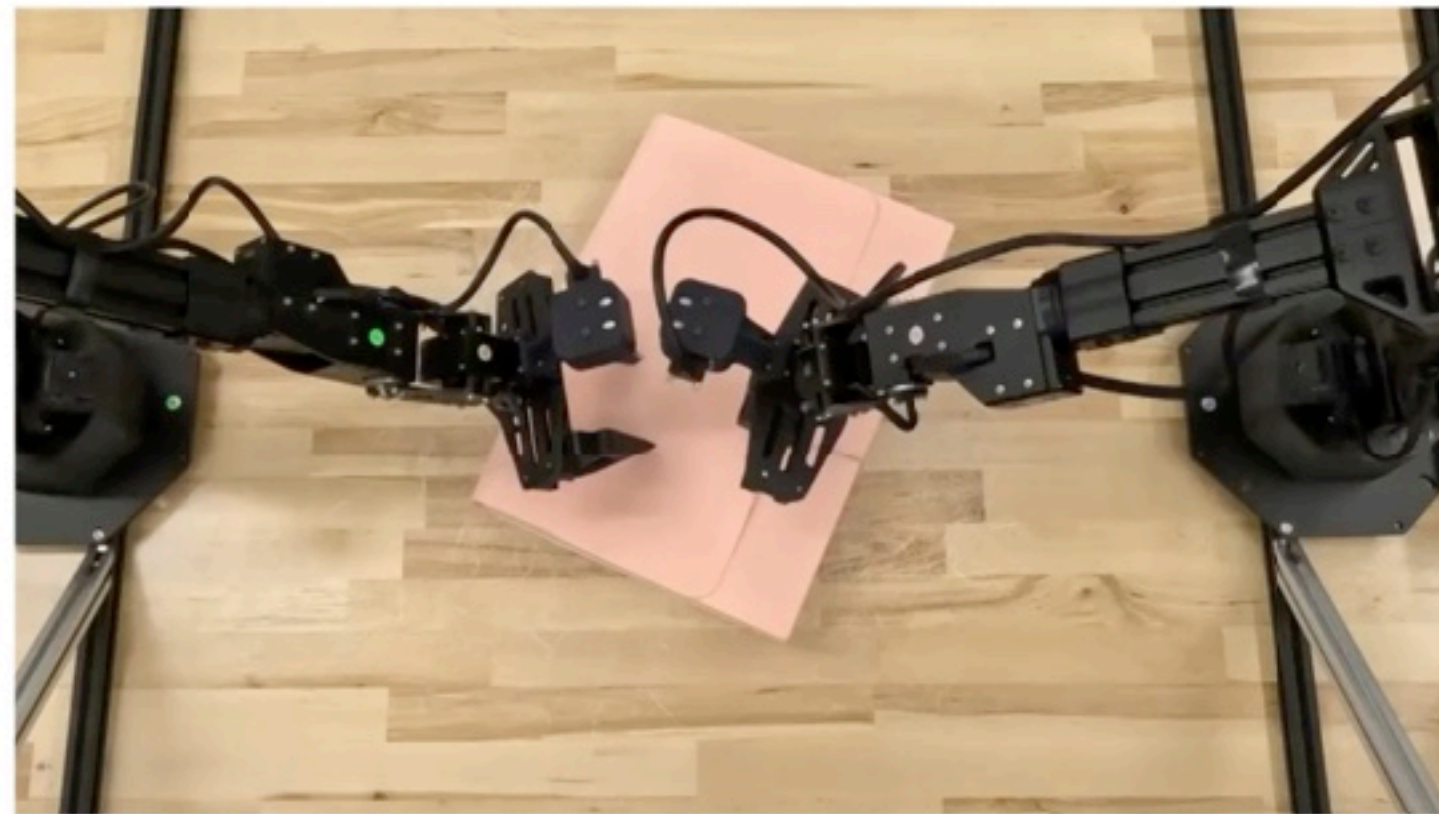
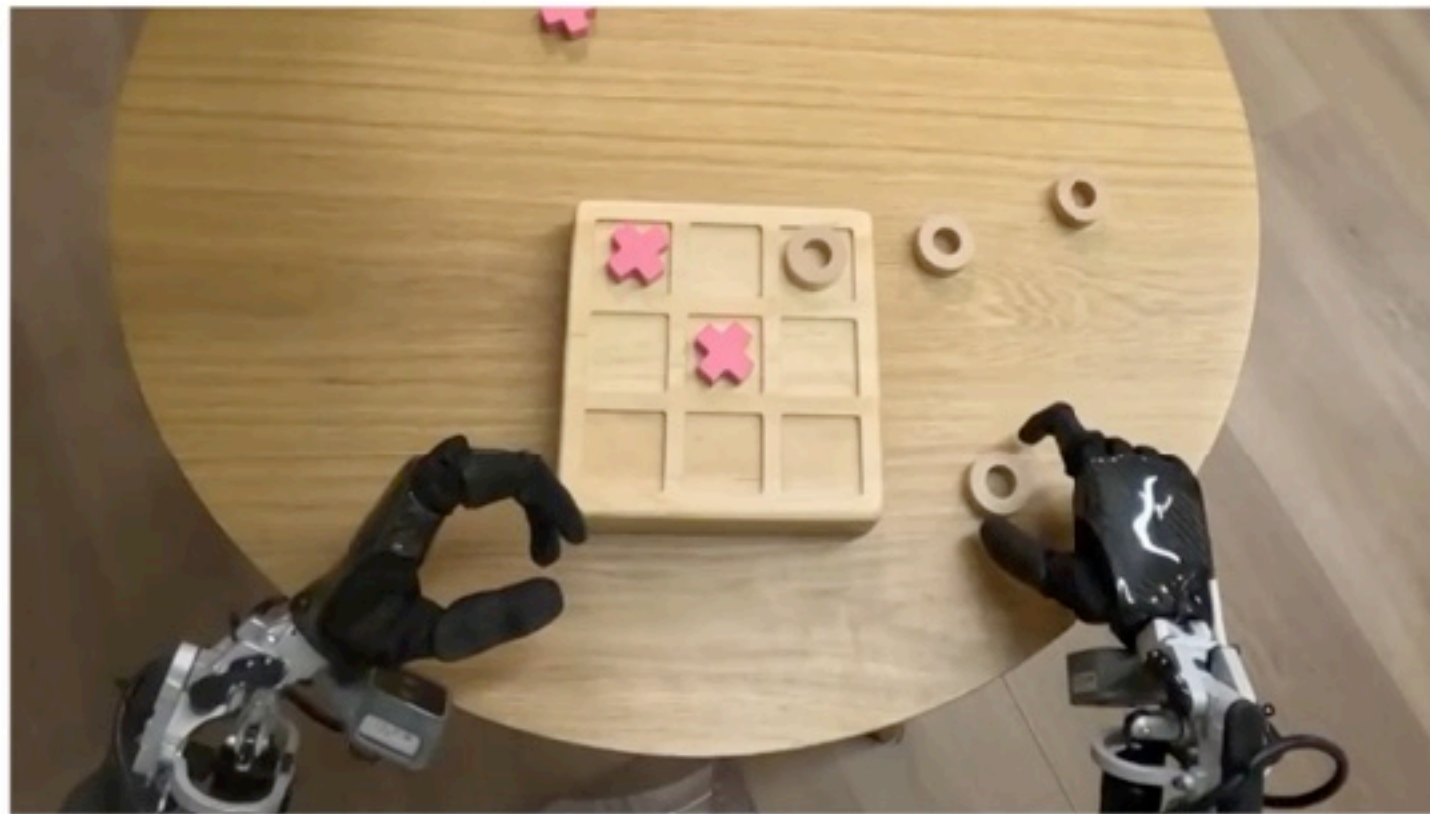
Video generation models



Runway Gen-4.5 (Dec 2025)



Seedance 2.5 (Jul 2026)



Apollo

Aloha2

Bi-arm Franka







Seoul World Model

Grounding World Simulation Models in a Real-World Metropolis

ECCV 2026 Oral

Junyoung Seo[†], Hyunwook Choi[†], Minkyung Kwon[‡], Jinhyeok Cho[‡], Siyoon Jil[‡], Gayoung Lee[‡], Junho Kim²

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Any questions?

Please don't hesitate to ask question for being a part of this discussion!